IT Alignment: An Annotated Bibliography

Article *in* Journal of Information Technology · October 2007

DOI: 10.1057/palgrave.jit.2000111 · Source: DBLP

Blaize Horner Reich
Simon Fraser University

Journal of Information Technology (2007) 1–81

palgrave-journals.com/jit

State of the Art

IT alignment: an annotated bibliography

Yolande E Chan[1], Blaize Horner Reich[2]

[1]School of Business, Queen's University, Queen's, Kingston, ON, Canada
[2]Faculty of Business Administration, Simon Fraser University, Vancouver, Canada

Correspondence:
YE Chan, Queen's School of Business, Queen's University, Kingston, ON, Canada K7L 3N6.
Tel: +1 613 533 2364;
Fax: +1 613 533 2321;
E-mail: ychan@business.queensu.ca

The authors invite readers to advise them of any inaccuracies, errors or omissions by commenting on the blog posting about this article published on the JIT Editor's Blog located at http://www.palgrave-joumals.com/jit/. From time to time, the authors will update the bibliography based on the information supplied by readers.

Abstract
We provide summaries of over 150 alignment articles. The information is intended to assist faculty and graduate students who are conducting IT alignment-related research. The findings presented should interest practitioners also. We hope that the article will facilitate the ongoing study and practice of IT alignment.
Journal of Information Technology advance online publication, 2 October 2007; doi:10.1057/palgrave.jit.2000111
Keywords: alignment; linkage; fit; methods; theories; findings

The following chart summarizes a literature review of many influential articles on information technology alignment.[1] It provides a snapshot of much of the literature to date in this area. Articles are reviewed in alphabetical order and are described based on their research method, theory/concept, and findings. These columns, however, are not mutually exclusive; column content overlaps at times. The research method indicates the type of study conducted by the authors, most commonly a conceptual paper, case study, survey, or other empirical study. Discussion in the theory or conceptual column elaborates on what the authors cite as the basis or starting point for their research. This column represents the article's foundation. The findings column is used to briefly present key points from the article, such as frameworks developed, hypotheses supported, significant relationships, best practices, and directions for future research. The summaries contained in this chart represent the interpretations of the articles produced by the researchers and their research assistants. Alternative interpretations or important subsidiary findings are welcomed by the authors. Should several be received, the *Journal of Information Technology* has agreed to post a revised version of this table on their website.

Acknowledgements
We thank research assistants Catherine Shea, Zorana Svedic, and Darius Tadaniewicz, and acknowledge the funding from the Social Sciences and Humanities Research Council of Canada that made their assistance possible.

Note
1 With the hundreds of articles available today on IT alignment, it was not possible to cite each article. We acknowledge that we have not recognized every study and apologize for any oversights. We also encourage any researcher whose contribution was overlooked or not appropriately presented to bring it to our attention.

Citation (Author, year)	Research method	Theory/concept	Findings
Al-Mashari, M. and Zairi, M. (2000). Creating a Fit Between BPR and IT Infrastructure: A Proposed Framework for Effective Implementation, *International Journal of Flexible Manufacturing Systems* **12**(4): 253–274.	Non-empirical study	This author proposed a framework for managing IT for effective business-process redesign (BPR) implementation, which is based on BPR principles, role of IT in BPR (Childe *et al*., 1996; Higgins, 1993; Lyons, 1997), and Henderson and Venkatraman's alignment model (1993).	The framework is based on the examination of BPR principles, the role of IT in BPR and the benefits of IT-enabled BPR. The authors presented various IT infrastructural considerations in BPR implementation and consolidated them into a number of elements found to be critical to successful outcomes of this process. Full benefits resulting from the introduction of enabling IT infrastructures for BPR initiatives will not be gained unless a fully integrated and balanced perspective is taken toward linking together all essential IT infrastructure management elements.
Allen, P.M. and Varga, L. (2006). A Co-Evolutionary Complex Systems Perspective on Information Systems, *Journal of Information Technology* **21**(4): 229–238.	Conceptual	The authors utilize complexity science and take a complex systems view of IS. The authors also discuss the concept of modeling.	Complexity science looks at the evolution of organizations resulting from the interaction of individuals, and explains the evolution of organization forms, structure, and capabilities. Coevolution occurs when two or more units interact to cause an evolutionary response. When applied to IS, complexity science tells us that our understanding of reality is contingent on the IS we use. IS are evolving systems of knowledge driven by individual values. Organizations coevolve over time due to interactions and thus, organizations are not stagnant. This presents a problem for alignment. Since an agent's view of reality is constantly changing, this leaves us with the problem of IS always lagging behind. The authors propose strategies to bridge this gap. They see role for the modeling process that would allow the IS to grow and develop based on the interaction with the individuals of the organization. Modeling capabilities would allow multi-agent representations of the organization and its environment in order to allow for more strategic exploration and decision making.
Avison, D., Jones, J., Powell, P. and Wilson, D. (2004). Using and Validating the Strategic Alignment Model, *Strategic Information Systems* **13**(3): 223–246.	Case study $N=55$ projects	This study derives its theoretical foundations from the strategic alignment model (SAM), developed by Henderson and Venkatraman (1989)	The paper proposes a practical tool designed to guide technology and business management alike towards the most productive utilization of technology resources. By examining the projects worked on over a previous period, management can retrospectively determine current alignment. It can then be used to monitor and track alignment and it can be used to pre-empt a change in strategy and implement a new alignment perspective by re-allocating project resources.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Baets, W. (1992). Aligning Information Systems with Business Strategy, *IS and Business Strategy Alignment* **1**(4): 205–213.	Case study $N=150$ managers	The Strategic Alignment Process of MacDonald (1991) and the Enterprise-wide Information Model of Parker, Benson and Trainor (1988) were used as a precursor to the introduction of an extended model for the strategic alignment process.	This paper develops an alternative approach to IS and business strategy alignment, based on the observation that corporate strategy is either unknown or un-adaptable once it is fixed. It argues, therefore, that *ex post* IS alignment will seldom be a success. The proposed approach is based on the current theoretical concepts in information management, linking them in a framework and integrating the process of corporate strategy definition with the process of IS strategy definition.
Baets, W.J. (1996). Some Empirical Evidence on is Strategy. Alignment in Banking, *Information & Management* **30**(4): 155–177.	Case study (including literature review, using the simulation tool, interviews and survey)	The theoretical foundation of this paper is the strategic alignment model (SAM) developed by Henderson and Venkatraman (1993).	The author concludes that IS Strategy Alignment is a process, including some business strategy, business organization, IS infrastructure and process, and IT strategy. The study demonstrates the importance and influence of mind sets on IS Strategy Alignment awareness. Awareness of the importance of banking problems has a significant influence on the understanding of the IS strategy and planning process. There is no strong and clear belief that IT can solve some specific banking problems, though the IS/IT issues are perceived important. This suggests that the awareness of IS/IT is of a theoretical nature and that managers have difficulties in translating the idea into the banking environment. Hence, the main problem in generating improved IS Strategy Alignment is a lack of overall banking industry knowledge, not skills, among banking managers. Corporate strategic planning is perceived as being important by managers for improved IS Strategy Alignment, and consequently IS strategic planning.
Baker, E.H. (2004, October 15, 2004). Leading Alignment, *CIO Insight* **1**(45) 19–20.	Survey $N=1100$ executives	Not applicable	This article highlights main survey findings (full results can be found in CIO Insight's third annual special issue on IT alignment, called 'Culture Clash'). Successful alignment between business strategy and IT requires a collaborative corporate culture and strong leadership. Corporate culture is defined as the norms of behavior and shared values that allow every organization's employees, at every level, to work together successfully toward a common goal.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Bassellier, G. and Benbasat, I. (2004). Business Competence of Information Technology Professionals: Conceptual development and influence on IT–business partnerships, *MIS Quarterly* **28**(4): 673–694.	Scale Development	This paper reviews the literature on the changing demands put on IT professionals, particularly increased demand on organization, business and interpersonal knowledge.	This research addresses the changing role of the IT in the overall organization. This paper addresses the notion of business competence in the IT professional. The authors define business competence in IT staff as 'the set of business and interpersonal knowledge and skills possessed by IT professional that enable them to understand the business domain, speak the language of business, and interact with their business partners.' A scale development was performed. Business competence is a function of organizational specific competence and interpersonal and management competence. Organization-specific competence was broken into elements of organization overview, organizational unit, organizational responsibility, and IT–business integration. Interpersonal and management competence was broken into knowledge networking, interpersonal communication, and leadership. Business competence was related to intentions to develop partnerships, and explains 28% of the variance.
Bassellier, G., Benbasat, I. and Reich, B.H. (2003). The Influence of Business Managers' IT Competence on Championing IT, *Information Systems Research* **14**(4): 317–336.	Survey $N=404$ business managers	Tests the conceptual model developed by Basselier *et al.* (2001). Also tests the proposition stated by Rockart *et al.* (1996) that line managers are more likely to assume leadership of IT when they have the appropriate education and training.	The results of this study show that IT knowledge and IT experience explain 34% of the variance in a manager's decision to champion an IT project. Shared knowledge between IT and business managers is established as a precondition to shared domain of knowledge, but this article makes its contribution to the literature by looking at one-half of the dyad – the business manager's IT knowledge. This study also makes a major contribution by breaking 'IT knowledge' into various scales.
Benbya, H. and McKelvey, B. (2006). Using Coevolutionary and Complexity Theories to Improve IS Alignment: A multi-level approach, *Journal of Information Technology* **21**(4): 284–298.	Conceptual	This authors base their model on the theory of co-evolution, McKelvey's (2004) 7 First Principles of Efficacious Adaptation, and scale free theory. The authors also review the literature on the emergent nature of strategy and the critiques of current alignment literature. Alignment is conceptualized as a dynamic, continuously adjusted process.	The authors propose a multilevel coevolutionary model of IT alignment. Alignment takes place at the individual level, operational level, and strategic level. At the individual level, IS infrastructure coevolves with end user needs. At the operational level, the IS department coevolves with the business. And at the strategic level, the IS strategy coevolves with the business strategy. Although these levels are distinct, the authors are more interested in the fractual structure. That is, as one moves down a level, a smaller yet otherwise identical

Citation (Author, year)	Research method	Theory/concept	Findings
			structure exists. This belief is consistent with scale-free theory. The enabling condition in this theory is the principles of adaptation and scale-free dynamics.
Bensaou, M. and Earl, M. J. (1998). The Right Mind-Set for Managing Information Technology (Japanese and American Methods), *Harvard Business Review* **76**(5): 119–129.	Conceptual	Not applicable	This article reviews some of the reasons that Japanese companies seem to be so successful at utilizing IT. The authors identify five principles or patterns that the Japanese follow when using IT and explain how these principles have led to success. The Japanese framing of IT includes strategic instinct, performance improvement, appropriate technology, organizational bonding, and human design. Strategic instinct differs from strategic alignment in the sense that Japanese management focus on operational goals as opposed to devising a strategic IT plan. Managers simply implement technologies that will enhance their capabilities. Performance improvement differs from the value for money approach traditionally taken in the West. Japanese investments are not made based on rational financial metrics, but rather based on their ability enhance operations. The traditional metrics often overlook projects that just do not meet the criteria. Appropriate technology differs from the technology for technology's sake mindset in North America. Often times this mindset leads to organizations adopting technology that is too advanced for their purposes. Japanese companies tend to adopt the appropriate technology for the task at hand, which may not be on the leading edge. Organizational bonding is emphasized in Japan as compared to IS user relations. A typical complaint in a North American firm is that the IT group does not know what managers actually do, and thus the system does not suit the organizational needs. In Japan, managers are rotated into the IT function and develop the systems on-site while working with line managers. Finally, the Japanese advocate for human design as opposed to system design. The Japanese develop their systems to fit the people who use them and do not try to force workers into systems that do not match their work patterns.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			The authors warn that exact of replication of the Japanese systems is not advisable. Rather, North American and European managers should use the principles outlined in this article in order to make more effective IT decisions.
Bergeron, F., Raymond, L. and Rivard, S. (2001). Fit in Strategic Information Technology Management Research: An empirical comparison of perspectives, *The International Journal of Management Science* **29**(2): 125–142.	Case study	The conceptual basis of this study is formed on the Venkatraman (1989) framework that comprises six different perspectives from which fit can be defined and studied, and on the findings of the study conducted by Chan (1997), which supported the moderation conceptualization of the fit.	Results obtained from applying and comparing the six fit perspectives demonstrates significant differences and confirms the need for conceptual and methodological rigor when applying contingency theory in strategic information technology management research. The study results reinforce Venkatraman's theory (1989) that different approaches to analyze fit might lead to different results. The authors noted this is the first empirical strategic IT management research to embrace the concept of fit in comprehensive and systematic manner.
Bergeron, F., Raymond, L. and Rivard, S. (2004). Ideal Patterns of Strategic Alignment and Business Performance, *Information & Management* **41**(8): 1003–1020.	Case study	The research model adopts a perspective of fit as gestalt (Miller, 1987). The alignment framework with four alignment domains (business, strategic, structural, IS) suggests six alignment types (Sabherwal and Hirschheim, 2001). This research also utilizes organizational contingency theory perspective.	The study results support the research proposition that confictual co-alignment patterns of business strategy, business structure, IT strategy, and IT structure will exhibit lower levels of business performance. However, using the fit between business dimensions and IT dimensions to explain performance is valid only if organizations have minimum thresholds on all four alignment domains, that is, are not systematically at the low end of the spectrum. Low-performance firms reveal a conflictual co-alignment pattern of business strategy, business structure, IT strategy and IT structure that distinguish them from other, higher performing firms. In order to achieve systems integration and increase performance when organizations face shifts in the business environment, managers must link changes in strategic choices across all four of the domains in a systematic way.
Bleistein, S. J., Cox, K. and Verner, J. (2006). Validating Strategic Alignment of Organizational it Requirements Using Goal Modeling and Problem Diagrams, *The Journal of Systems and Software* **79**(3): 362–378.	Theoretical development and case study (one site: Seven-Eleven Japan)	The model was developed using VMOST analysis (Sondhi, 1999) and the BRG-Model (Kolber *et al*., 2000) in conjunction with goal modeling. They structured the model according to Jackson's problem diagram framework to encompass both	This paper presents an integrated approach to requirements engineering for organizational IT alignment. The motivation for the research was to highlight the fact that alignment research tends to ignore connections to systems requirements. The authors illustrated an explicit link between system requirements and the objectives of business strategy.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
		strategy and system requirements. VMOST is used for analyzing and deconstructing business strategy into component parts. BRG provides rules for relating each of the components to business strategy.	The goal model provides a mechanism for verifying alignment as it enables explicit connections to requirements at adjacent levels in terms of super goals and sub goal. Thus, context diagrams helps ensure that requirements are consistent with business and system context, while the goal model helps ensure that system requirements achieve business objectives. At the same time, integrating goal modeling with problem diagrams helps improve manageability of requirements analysis of complex systems. Using this model, lower level technical requirements could be linked to higher level organization strategy.
Bradach, J. (1996). *Organizational Alignment: The 7-S Model No. 9-497-045)*, Cambridge, MA: Harvard Business School Note.	Conceptual	Not applicable	This note describes the 7-S model to organizational alignment. It serves as a framework for managers to assess and improve their organization's level of alignment. The 7-S model includes strategy, structure, systems, style, shared values, skills, and staffing. All elements are related and a change in one will impact all others. The 7-S model is embedded in a given environment. The author hopes that this model will help managers identify opportunities and manage change and alignment more effectively.
Breu, K. and Peppard, J. (2003). Useful Knowledge for Information Systems Practice: The contribution of the participatory paradigm, *Journal of Information Technology* **18**(3): 177–193.	Research Methods	This paper is a reflection of the research methods utilized over a longitudinal study. This paper utilizes the interventionalist form of research and the participatory framework.	The authors and practitioner colleagues have worked on a decade long project on IT alignment. In this paper they share their views on the participatory worldview and attempt to suggest a system of knowledge production that is complementary to existing modes of IS inquiry, recognizing the fact that there will always be natural limitations associated with different ways of knowing. The intention in this paper is to demonstrate that philosophical foundations for interventionist research strategies do exist. Interventionist approaches are defined in this paper so as to include any form of inquiry that is committed to the implementation of research results, as, for instance, with action research, action science, cooperative inquiry and participatory inquiry. It is advocated that scholars focus their efforts more on adopting and developing research approaches that create useful IS knowledge rather than theorizing over the commensurability of rigor and relevance, without proposing a viable alternative.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Broadbent, M. and Kitzis, E. (2005). Interweaving Business-Driven IT Strategy and Execution: Four foundation factors, *Ivey Business Journal* **69**(3): 1–6.	Conceptual	Not applicable	This article discusses four factors that the authors deem to be important for building business-IT linkages. The first factor is a CIO who provides leadership on both the demand and supply sides of their role. The demand side means that the CIO must actively provide a vision of IT to support business strategy. On the supply side, the CIO must deliver cost effective IS. The second factor is that an executive team must take the time to develop informed expectations for an IT-enabled enterprise. The third factor is that clear and appropriate demand-side IT governance is needed. This will ensure that there are clear expectations of who's responsible for what so that faster and more effective decisions can be made. The final factor is that an organization must take an IT portfolio management approach.
Broadbent, M. and Weill, P. (1993). Improving Business and Information Strategy Alignment: Learning from the Banking Industry, *IBM Systems Journal* **32**(1): 162–179.	Case study $N=4$ banks	Evidence for alignment was sought in the use of information or information technology, or both, which provided a comparative advantage to an organization over its competitors. The conceptual basis for this approach is the realized strategy framework of Mintzberg (1978).	This paper has identified organizational practices that contribute to alignment and has outlined a series of questions with which managers can review the practices of their firm. Different organizational practices among four banks provided grounded evidence for 15 propositions about alignment. These propositions were grouped into an alignment model that emphasizes the interdependence of firm-wide and IS/T factors. A key finding produced by this paper was that the most effective use of IT/IS occurred when it was managed by someone close to the business needs (i.e., a non-IS manager).
Broadbent, M., Weill, P., O'Brien, T. and Neo, B. S. (1996) Firm Context and Patterns of IT Infrastructure Capability. Unpublished manuscript.	Case study $N=26$ large business units	The authors developed and presented the concept of IT infrastructure capability. IT infrastructure capability is comprised of IT Infrastructure Services and IT reach and range. IT infrastructure capabilities are a function of firm context (industry, marketplace volatility, BU synergies, and strategy formation processes).	This study shows that IT infrastructure capabilities are more complex than that originally operationalized. IT infrastructure capabilities is contingent on the level of information intensity and strategic focus. At lower levels, there are 5 core services: communications management, standards management, IT education management, security and data management. However, at higher levels of information intensity and strategic focus, services such as applications management, IT management, and R&D gain importance. The study demonstrated a strong association between firms with higher IT infrastructure

Citation (Author, year)	Research method	Theory/concept	Findings
			capabilities and firms that value BU synergies, integrate IT into overall business planning processes, and have firm-wide specific practices in place to ensure strategic coordination.
Brown, C.V. and Magill, S.L. (1994). Alignment of the IS Functions with the Enterprise: Toward a model of antecedents, *MIS Quarterly* **18**(4): 371–403.	Case study $N=6$ for profit firms	The article expands on the concept of IS organizational designs (centralization, decentralization, hybrid). It also examines the patterns of internal and external antecedents for explaining these different choices. The initial conceptual framework was developed and then evaluated.	This study explains a firm's IS organization design decision for a decentralized, centralized, or hybrid locus of responsibility from an expanded set of environmental, overall organizational, and IS-specific antecedents as well as a larger concept of organizational alignment. The study findings provide support for the two related trends in IS alignment reported in the literature: (1) both computer and telecommunications operations are most commonly regarded as utility functions, consolidated under a central IS unit, and (2) decentralization solutions have primarily been directed at the management of the use of technology. The study also supports the assumption that different firms in the same industry choose different IS organization designs. Finally, this article provides the evidence that centralized, decentralized, hybrid, and split IS structures can be effective. The key is to choose the appropriate configuration suitable for the company.
Burn, J.M. (1993). Information Systems Strategies and the Management of Organizational Change – A Strategic Alignment Model, *Journal of Information Technology* **8**(4): 205–216	Case study $N=56$ organizations	The development of the strategic alignment framework was based on IS literature and practice over the last 30 years regarding the organizational strategy and information systems strategy formulations. Six analogous phases were identified that link both of these strategies together.	A theoretical organizational cultural audit (OCA) framework is developed to examine the relationships between organizational and IS strategy formulation processes. Results of the analysis support the theory that different stages of growth in the use and development of IS require different approaches to strategy, and that different approaches to strategy are favored by different organizational configurations. As companies evolve through stages of the strategic alignment model, they experience rapid periods of change. The flow through the alignment model can be predicted from these rapid periods of change.
Burn, J.M. (1996). IS Innovation and Organizational Alignment – A Professional Juggling act, *Journal of Information Technology* **11**(1): 3–12.	Longitudinal study $N=168$ organizational studies	The framework used in the research study combines theories related to organizational behavior with those related to IS strategies in a comprehensive methodology and is	A strategic alignment model has been shown to exist at the functional level (internal alignment) and at the strategic level (external alignment) which supports a dynamic model of change and the need to adopt a lead–lag model of IS strategy. A cyclical model of IS

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
		known as the organizational cultural audit (OCA) (Burn, 1993).	innovation is proposed which can assist the innovative organization to adapt effectively and efficiently to the impact of IS on organizational change. This model is proposed as a *'r' evolutionary* concept of IS planning.
Burn, J.M. (1997). A Professional Balancing act – Walking the tightrope of strategic alignment, in C. Sauer and P.W. Yetton (eds.) *Steps to the Future – Fresh Thinking on the Management of IT-Based Organizational Transformation,* 1st edn., San Francisco: Jossey-Bass Publishers, pp. 55–88.	Longitudinal study $N = 250$ organizations	The author develops a dynamic model of alignment, Lead–Lag Model, based on the Organizational Culture Audit (OCA) (Burn, 1993).	The author makes three main conclusions about the alignment process: • Alignment is not a one-time activity but a constant balancing act between a lead or lag strategy. • Alignment through lead–lag is typified by cycles of change, and these cycles tend to be specific to particular organizational types and particular industries. • Knowing these cycles of change and your organizational position in relation to them will facilitate management of the alignment process.
Burn, J.M. and Szeto, C. (2000). A Comparison of the Views of Business and IT Management on Success Factors for Strategic Alignment, *Information & Management* **37**(4): 197–216.	Case study	The authors use the strategic alignment model (Henderson and Venkatraman, 1989, 1993). It assumes existence of four perspectives of strategy alignment, (strategy execution, technology transformation, competitive potential and service level). The questionnaire utilized in this paper is based on surveys conducted by Galliers *et al.* (1994) as well as Henderson and Venkatraman.	The research study assumes that effective alignment of IT and business strategies can be achieved by means of strategic information systems planning (ISIP). The survey results indicate that the perspectives of IT managers and business managers are remarkably consistent with both groups selecting technology transformation as their overall perspective and business strategy as the driver. However, despite previous extensive validation of the alignment model, this research indicates that the model does not accord with reality. The authors suggest that companies, which perceive future strategy becoming technology dependent, should follow the suit of leaders, who concentrated on technology transformation and competitive potential.
Byrd, A., Lewis, B.R. and Bryan, R.W. (2006). The Leveraging Influence of Strategic Alignment on IT Investment: An empirical examination, *Information & Management* **43**(3): 308–321.	Empirical study, $n = 84$ pairs of plant managers and IT managers in small and medium manufacturing plants	The authors tested competing theories of strategic alignment. Two focused on alignment in the planning process of business and IT (coordination and integration). The other two focused on the realized or outcomes of alignment (matching and moderating). The authors also proposed a research model that	This paper presents an empirical examination of the moderating affect of strategic alignment on the relationship between IT investment and firm performance. Results show a synergistic coupling between strategic alignment and IT investment with firm performance. The real value of strategic alignment is in leveraging the firm's IT investment. Firms with poor alignment can increase profits and revenues without investing more in IT, but rather by better

Citation (Author, year)	Research method	Theory/concept	Findings
		strategic alignment has a direct impact on performance as a moderator between IT investment and business performance.	aligning business and IT strategies. Those organizations that have good alignment can invest more in IT with some confidence that it will be leveraged substantially. Strategic alignment of IS and business strategy created additional positive effects on revenue per employee above and beyond the use of IT investment and strategic alignment individually. The process of *integration* alignment was shown to be not necessarily useful. This indicates that organizations may still face problems when trying to develop business and IT plans simultaneously. This may be because the gap between IT managers and management in SME may be too pronounced.
Campbell, B. (2005). *Alignment: Resolving Ambiguity within Bounded Choices. PACIS 2005*, Bangkok: Thailand.	Multiple Focus Groups and Interviews	This study uses grounded theory. The grounded theory was informed by relativist ontology and constructionist epistemology.	The essence of the theory is that business managers are confronted with a dilemma when most formal business strategies are presented to them. They implement these strategies, resolving their own ambiguity, according to: • Their ability to understand the problem or concept encapsulated within the strategy (Locus of Comprehension) and • Their freedom to actually make choices (Locus of Control). There have been consistent calls within the IS alignment literature to improve communications with senior business managers, gain senior management commitment to IS strategies, and increase business knowledge in IS managers, among others. However, the current research and substantive theory indicates that for many IS managers and groups these recommendations/requirements may be totally beyond the bounds of possibility due to the locus of comprehension and the locus of control.
Campbell, B., Kay, R. and Avison, D. (2005). Strategic Alignment: A Practitioner's Perspective, *Journal of Enterprise Information Management* **18**(5/6): 653–664.	Focus group $N=6$ managers	Exploratory study using grounded theory. Analyzed qualitative data.	This research indicates that the major concern of IS managers when considering alignment is the *ambiguity* surrounding the differences between espoused strategies, strategies in use and the actions of business managers that support self-interest. In coping

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			with this ambiguity, IS managers at all levels of an organization adopt one of two coping responses: Technological or Collaborative. The choice of response is bounded by various factors making up the Locus of Comprehension and Locus of Control. Each response has certain characteristics that affect both the level and target of alignment. Once a coping response has been established it then becomes extremely difficult to change. This study also produced a very useful model for viewing alignment.
Chan, Y.E. (2002). Why Haven't we Mastered Alignment?: The Importance of the Informal Organization Structure, *MIS Quarterly Executive* **1**(2): 97–112.	Case study of eight organizations were considered 'top performing'. Five companies were located in Canada and three were in the US.	Examines the importance of strategic alignment and structural alignment.	In studying these top performers, it was found that alignment research and practice are generally in sync, with the exception of the need to document the IS and business plans and having the IT personnel involved in product and service development. However, there was no one right way to align; each organization aligned in different ways. Strategic alignment mattered more than structural IS alignment. The IS structures had to be flexible and these structures were primarily a means to an end. Mostly importantly, it was found, that the informal organizational is more important than previously thought in achieving alignment. This informal structure is reinforced by the culture. Future research is needed to fully understand the alignment benefits of the informal structure.
Chan, Y.E. and Huff, S.L. (1992). Strategy: An Information Systems Research Perspective, *Journal of Strategic Information Systems* **1**(4): 191–204.	Conceptual paper/ Literature Review	This article is intended to provide an overview of business strategy research findings and recommendations that are especially relevant to IS researchers, focusing on key conceptual and measurement issues in three areas: competitive strategy, strategic fit, and business performance.	The authors outlined the following recommendations as to how to research IS strategy: • Both strategy content and strategy process issues may need to be considered when conducting strategy-related research. • Researchers may focus on strategy formulation and/or implementation issues. • Researchers must decide whether to focus on company goals as well as means. • It is important for researchers to distinguish between business unit strategy and corporate strategy. • When studying competitive strategy, researchers should consider working with firms operating in or concentrated in single industries.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			• Researchers should utilize CEOs and top ranking officials as information sources when assessing competitive strategy. • To obtain a rich and detailed understanding of strategy from multiple viewpoints, consider case studies or historical approaches. • If simplicity of measurement is important, and categorical data are acceptable, consider using a typology to assess strategy. • If precision is important, and many firms are to be studied, employ either a taxonomy or comparative measurement approach. • Alignment – or fit – is fundamental to the notion of strategy. • Researchers should include business performance as a key variable in strategy-IT studies. • In IS-strategy studies, it is important also to gather data on key company and environmental contingency factors. • The relationships between strategy and performance should be studied within and not across industries whenever possible. • It is important to identify major strategic groups within industries studied. • Researchers should use multiple informants to assess business performance and/or report the level of respondent agreement. • Whenever possible, utilize subjective and objective performance data from both primary and secondary sources.
Chan, Y.E. and Huff, S.L. (1993). Investigating Information Systems Strategic Alignment, Proceedings of the Fourteenth International Conference on Information Systems, Orlando, FL. 345–363.	Instrument Development and Validation	The authors introduced the conceptual model of IS strategic orientation based on the STROBE model (Venkatraman, 1985, 1989) and alternative calculations for IS strategic alignment.	The authors developed the STROIS instrument, a parallel instrument to STROBE, in order to assess the alignment of IS to the business strategy. This study focused on two models for strategic alignment: matching and moderation. The moderation approach received most empirical support. Key findings were that IS strategic alignment is consistently related to perceived IS effectiveness and that alignment is positively related to innovation, financial performance (weak support), and market growth (weak support). This study advises practitioners that system support is most effective when it is focused on a few priority areas.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Chan, Y.E., Sabherwal, R. and Thatcher, J.B. (2006). Antecedents and Outcomes of Strategic IS Alignment: An Empirical Investigation, *IEEE Transactions on Engineering Management* **51**(3): 27–47.	Three separate surveys were used. Surveys 1 and 2 looked at Canadian and US businesses listed in Dunn & Bradstreet directories and surveyed the CEO and CIO about business and IT strategies. Survey 3 looked at academic institutions that were in the top 650 in the US in terms of enrollment.	Using a contingency model of strategic IT alignment, the authors examined antecedents: shared domain knowledge, planning sophistication, prior IS success, environmental uncertainty and organizational size.	This paper tests an integrated empirical model of alignment. Specifically, it looks at antecedents to alignment in different business industries (public and private) and in businesses with different strategies (Defender, Analyzer, Prospector). This study provides empirical support for the popular argument that alignment improves organizational performance, but it is argued that not all firms are equally well served by allocating scarce resources to improve IS alignment.
Chan, Y.E., Huff, S.L., Barclay, D.W. and Copeland, D.G. (1997). Business Strategic Orientation, Information Systems Strategic Orientation, and Strategic Alignment, *Information Systems Research* **8**(2): 125–150.	Mail survey	The foundation for this study is an extension of Venkatraman's (1985, 1989) conceptualizations of fit and STROBE. This extension includes IS in his model.	This study emphasizes that creating an IS strategy is a first step; a second step – realizing the strategy – is also critical. Companies that appear to perform best are companies in which there is alignment between business strategy and *realized* IS strategy. This alignment was linked to IS effectiveness and business performance. The study suggested that the IS strategic orientation has three core dimensions: IS support for Anticipation, IS support for Analysis, and IS support for Action. This study also makes a contribution by extending business strategy and alignment measures. Specifically, STROEPIS was developed which measures the realized IS strategy.
Chan, Y.E., Huff, S.L. and Copeland, D.G. (1998). Assessing Realized Information Systems Strategy, *Strategic Information Systems* **6**(4): 273–298.	Instrument development and survey	The study is based on the notion of strategic orientation STROBE (Venkatraman, 1985, 1989) and STROIS (Chan and Huff, 1992). The authors distinguish between intended strategy and realized strategy (Mintzberg, 1978). A scale assessing realized IS strategy is developed.	This study attempts to answer the questions: 'How does one go about understanding and assessing *realized* IS strategy?' This question distinguishes between two key factors of an organization: its priorities and its capabilities. Just because an organization has a documented business strategy or IS strategy does not mean that it has the capabilities to implement the planned strategies. The authors also cite the need to quantitatively study IS strategy and thus develop a measure. The developed measure factored into four components: IS to support company future orientation ('Anticipation'), IS to support company defensive orientation ('Armour'), IS to support company analysis and risk aversion ('Analysis'), and IS to support company action ('Action'). This measure provides a relatively complete picture of realized IS strategy for the firms examined.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Chan, Y.E. (2001). Information Systems Strategy, Structure and Alignment, in R. Papp (ed.) *Strategic Information Technology: Opportunities for Competitive Advantage,* 1st edn., Hershey, PA: Idea Group Publishing, pp. 56–81.	Survey $N = 67$ CEO–CIO pairs	The study is focused on strategy, structure, alignment and performance at the business and IS levels. IS alignment is modeled using two dimensions, namely strategic and structural alignment, which are related to the previous research in the field (Henderson and Venkatraman, 1993; Luftman, 1996; Papp, 1998).	The study reveals several relationships among key constructs. Among other things, it indicates the complexity of IS alignment, which involves two, related, yet distinct dimensions: strategic and structural. Both IS strategic alignment and IS structural alignment proved to be very important from a performance point of view. The quality of the information product is clearly linked to financial performance. A strong link is also revealed between IS performance and company innovation and company reputation. The study presents a set of alignment factors that should be monitored and emphasizes the importance of the synergy between business and IS strategic orientation, rather than a simple match between them.
Choe, J. (2003). The Effect of Environmental Uncertainty and Strategic Applications of IS on a Firm's Performance, *Information & Management* **40**(4): 257–268.	Survey $N = 70$ organizations	The authors based their theoretical foundations on previous research on alignment, environmental uncertainty, and resource dependency theory. This study considers perceived environmental uncertainty (PEUN) and alignment factors as key research variables that may directly or indirectly relate to strategic applications.	The article examines the relationships among perceived environmental uncertainty (PEUN), the level of the strategic applications, and the facilitators of IS strategic alignment. Using a systems approach, some ways that these relationships impact the performance of a firm are shown. PEUN has an indirect effect on IS strategic applications through the facilitators of alignment. An uncertain environment, a high level of strategic applications and well-arranged facilitators of alignment can contribute more to the improvement of performance than in a less uncertain or stable environment. Therefore, the benefits of a high level of strategic applications can still be realized in a competitive and uncertain environment. However, when the environment is stable (low PEUN), an excessively low level of strategic applications may decrease a firm's performance.
Chung, S.H., Rainer Jr., R. Kelly and Lewis, B. R. (2003). The Impact of Information Technology Infrastructure Flexibility on Strategic Alignment and Applications Implementations, *Communications of the Association for Information Systems* **11**(11): 191–206.	Case study	The authors build the theoretical foundations by reviewing definitions of IT infrastructure and its components, and by defining the concept of IT infrastructure flexibility and its relationships to strategic IT-business alignment, and to applications implementation in the organization using four previously	This study looks at the concept of IT infrastructure flexibility. The results reveal that three components of IT infrastructure flexibility, namely connectivity, modularity, and IT personnel, contribute to strategic alignment. This study reveals that compatibility did not have a significant impact on strategic IT–business alignment. IT infrastructure, in order to facilitate organizational responses to dynamic environments, must be flexible and supported by the strong

Citation (Author, year)	Research method	Theory/concept	Findings
		identified measures of IT infrastructure flexibility: modularity, compatibility, connectivity, and IT personnel skills (Duncan, 1995; Byrd and Turner, 2000).	alignment of IT strategy with the organizational strategy. All four components of IT infrastructure flexibility impact the extent of applications implementation in an organization significantly and positively.
Ciborra, C.U. (1997). De Profundis? Deconstructing the Concept of Strategic Alignment, *Scandinavian Journal of Information Systems* **9**(1): 57–82.	Conceptual paper	This paper engages an inquiry pertaining to both the notion of strategic alignment in business and the business of conducting research, publishing, and in general, doing 'management science' according to the prevailing business school model.	The paper's post mortem reflection on the managerial research programme on alignment has pointed out several new tracks: • A style of research that does not estrange us from the worldly existence of people at work and their small and big 'dramas' • The ensuing urgency to go back to the facts themselves, putting into brackets received concepts such as strategy, technology, and in general the power of models and representation • A new language to talk about the interaction between strategy and technology, including the terms of care, hospitality and cultivation • A series of new perspectives on technology, that go beyond the notion of technology as a means to achieve an end • An enlarged notion of alignment within an hybrid network of semiautonomous actors
Ciborra, C.U. (1998). Crisis and Foundations: An Inquiry into the Nature and Limits of Models and Methods in the Information Systems Discipline, *Journal of Strategic Information Systems* **7**(1): 5–16.	Conceptual paper	The author indicates that the reason for the present success of IS discipline is due to factors that are somehow extraneous to it, i.e. the Internet and the strategic use of information technology. Both innovations stem from initiatives that have grown outside academia, and especially outside the focus of academic teaching and research in information systems: methodologies for systems analysis, design and development.	The author believes that there is a need to go back to the world of practice to find the foundations of a new style of information systems teaching and research. The paper traced the origins of the crisis in the IS discipline back to the dominating style of research based on the scientific paradigm. In trying to closely mimic the Galilean paradigm, the so influential style of research that is not preoccupied with thinking. This is needed because science is too concerned with observation, measurement and calculation, not strictly with thinking. The author focuses his attention on the crisis generated by the scientific method applied to the design of human affairs and computer systems. Newer systems, such as strategic information systems, Internet, and in general the emergence of global IT infrastructures, all seem to suggest that technology

Citation (Author, year)	Research method	Theory/concept	Findings
			may require us today to speak another language, less formal and structured and more fragmented. The structured methodologies are too naïve and do not capture the intricacies of everyday life, nor the next challenge for ubiquitous and invisible computing; thus, old methodologies should be dropped.
Cragg, P., King, M. and Hassin, H. (2002). IT Alignment and Firm Performance in Small Manufacturing Firms, *Strategic Information Systems* **11**(2): 109–132.	Survey $N=256$ organizations	This study focused on measuring the alignment of business strategy and IT strategy (ITS) among small UK manufacturing firms and then investigated the link between alignment and performance.	The results indicated that a significant proportion of small firms had achieved high IT alignment. Furthermore, the group of small firms with high IT alignment had achieved better organizational performance than firms with low IT alignment. This is consistent with findings in large firms and opens up possibilities for further study of IT alignment in small firms. However, the authors do not know how this alignment was achieved. A quarter of the sample (26%) had formalized their ITS, while 68% of the sample had a written business plan. This supports the notion that IT planning does exist in small firms, but much of it is carried out informally.
Croteau, A. and Bergeron, F. (2001). An Information Technology trilogy: Business Strategy, Technological Deployment and Organizational Performance, *Journal of Strategic Information Systems* **10**(2): 77–99.	Case study $N=223$ Canadian firms	The construct of the research is based on: • Classification of business strategies (Miles and Snow, 1978); • Technological deployment concept based on five conceptual frameworks addressing the strategic concept (McFarland *et al*., 1983; Porter and Millar, 1985; Das *et al*., 1991; Henderson and Venkatraman, 1999; Bergeron and Reymond, 1995); • Organizational performance concept.	This study attempts to identify matches of technological strategies to various strategic orientations that best support the business strategy. The study reveals a positive link between strategic activities and organizational performance for organizations with prospector strategies, and a negative link for organizations with reactor strategies – as indicated in the first hypothesis. The relationship between strategic activities and different profiles of technological deployment, as described in the second hypothesis, is accepted for prospector, analyzer and defender strategies. The existence of positive relationship between strategic activities, specific profiles of technological deployment and organizational performance, as presented in the third hypothesis, is accepted for prospector and analyzer strategies. The study results indicate that organizations can enhance performance by supporting prospector or analyzer strategies and deploying IT accordingly.
Das, S.R., Zohar, S.A. and Barkentine, M.E. (1991). Integrating the Content	Conceptual	This research looks at two types of strategic MIS planning, namely MIS	This paper presents an integrated framework that links strategic MIS planning and business strategy, and

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
and Process Of Strategic MIS Planning with Competitive Strategy, *Decision Sciences* **22**(2): 953–985.		strategic planning methodologies and MIS implications for the phases of competitive strategy. This paper addresses the flaw of these two planning types, namely their lack of focus on linking the content and the process. The authors also utilize the concept of fit and strategic orientation (Miles and Snow, 1978).	relates it to competitive advantage and company performance. The authors state that effective strategic MIS planning must be coordinated on both the content and the process. The content of MIS strategic planning involves distinctive competence, IS technology, systems design and development, and infrastructure. The process of strategic MIS planning relates to issues of formality, scope, participation, influence, and coordination. MIS planning process is hypothesized to be different depending on the strategic orientation. Defenders are hypothesized have a highly formalized, top-down approach to MIS planning. Analyzers are thought to have a moderately controlled, top-down approach to planning. Prospectors are thought to have an informal, flexible, bottom up approach to planning. In terms of the content of the MIS plans, strategic orientations are hypothesized to have certain technologies to match their overall business strategy. Four overall propositions are developed. First, the content and process of strategic MIS planning are a source of competitive advantage, and ultimately, superior performance. Second, the fit between an organization's strategic orientation and MIS planning is positively associated with company financial performance. Third, the fit among the dimensions of a firm's strategic MIS planning within a particular competitive strategy is positively associated with firm performance. Finally, for each strategic orientation, failure to match content and process factors will be associated with diminished financial performance. The authors recommend that practitioners view MIS strategic planning at a higher level and ensure that it matches the firm's overall strategic orientation. The authors recommend that researchers engage in empirical research to further examine the steps in achieving alignment.
Day, J.G. (1996). An Executive's Guide to Measuring I/S, *Strategy & Leadership* **24**(5): 39–41.	Conceptual paper	The three sets of measurements presented here have been applied successfully in actual practice to help managers and executives get a handle on the most important elements of I/S performance.	Two factors often conspire to blacken the image of Information Services in many organizations: (1) end users tend to be removed from direct control and measurement of the IS function, and (2) they tend to see IS as overhead and administration, an uncontrollable cost with no appreciable advantage.

Citation (Author, year)	Research method	Theory/concept	Findings
			The measurements fall into three categories: (1) *Key measurements* that benchmark the IS function against industry averages and let you know whether you are performing above or below the norm; (2) *Alignment measurements* that help determine whether the IS function is operating effectively doing the right things in support of the company's business goals at the strategic level; (3) *Operational measurements* that help determine whether specific IS functions are operating efficiently doing things right, with tactical effectiveness for customer departments. Alignment Measurements: Alignment indices are used to determine how effectively the I/S function is supporting the company's overall goals at the strategic level: • Alignment Index (Al) – Business objectives are usually communicated by vision or mission statements. At a qualitative level, a simple comparison of the IS activities to those stated business goals can be very instructive. In making the comparison, assign a percentage value based on a scale of 1 to 100 that represents the subjective evaluation of how well aligned each activity is with the business objective it supposedly serves. • Effectiveness Acid Test (EAT) – A direct comparison between the proportion of IS expenditures devoted to specific P&L line activities and the volume of each activity expressed as a percentage of sales. Those two figures should be similar in magnitude over the long haul. The EAT is based on the postulate that the volume of a given activity, the resources needed to support it, and the requisite level of IS support are all reflected in the cost. The EAT is extremely time sensitive.
Deans, P.C. and Ricks, D.A. (1993). An Agenda for Research Linking Information Systems and International Business: Theory, methodology, and application,	Conceptual paper	A model developed by Deans and Ricks (1991) extends the traditional domestically oriented IS research frameworks to include variables specific to the international environment. In this paper, their	This article attempts to discuss the international business (IB) research that is relevant and applicable to the information systems researcher. There is a need for an international IS research agenda that builds on relevant past foundations and reflects the realities of a new and evolving business environment.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Journal of Global Information Management **1**(1): 6–20.		model distinguishes between foreign, international, and domestic components of the firm's external environment.	This article provides the IS reader with a structured framework from which to proceed in developing a research focus that interfaces the IS and IB research traditions. The authors state that a research agenda that provides a starting point from which to build through future research efforts is presented. Interdisciplinary work will be necessary to adequately address the complex issues that characterize the IS/IB interface. Collaboration among academicians from both IS and IB will add richness to the research under study. The IS/IB interface offers the researcher a challenging and intellectually stimulating area of academic investigation where the opportunities for research are immense.
Drazin, R. and van de ven, Andrew H. (1985). Alternative Forms of Fit in Contingency Theory, *Administrative Science Quarterly* **30**(4): 514–539.	Empirical	Authors present and evaluate the structural contingency theories, investigating fit between technology and organizational structure.	After evaluating theories, the authors presented empirical tests of three approaches to fit by focusing on the contingency theory. Results indicate that the structure and process choices for a particular organizational level is constrained and limited by design criteria required by macro-organizational levels. Findings support a managerial selection of fit, which has important implications for understanding the patterns of fit in analyzing the contingency theory. Findings also suggest that explaining the performance of organizational units requires a more sophisticated approach to the contingency theory than earlier studies suggest.
D'Souza, D. and Mukherjee, D. (2004). Overcoming the Challenges of Aligning IT with Business, *Information Strategy: The Executive's Journal* **20**(2): 23–31.	Conceptual	The authors base their arguments around four macro trends: the diminishing value of tangible assets; the rise of the knowledge worker; the transient nature of competitive advantage; and the unpredictability of market upheavals.	This article is inspired by the following paradox: Why does the empirical evidence not support the theoretical arguments for alignment? This article asks practitioners to rethink the IT-alignment paradigm. The authors suggest many reasons for potential failures. First, an over-reliance on IT is stated. That is, practitioners may adopt IT to be strategy as opposed to complement strategy. The authors state that IT is a necessary, but not sufficient condition for competitiveness. Secondly, the temporal nature of IT is addressed. IT is a long-term solution, but expectations are rather short term. Third, the notion of change is addressed. Any new system inspires change that is difficult to manage.

Citation (Author, year)	Research method	Theory/concept	Findings
			The authors suggest the following to practitioners: recognize the detrimental effects of resilient mental models and give yourself ample time to study, understand, and alter them; go to great lengths to clearly define and measure IT investment; and view IT as a partner, not as a subordinate.
Duncan, N.B. (1995). Capturing Flexibility of Information Technology Infrastructure: A study of resource characteristics and their measure, *Journal of Management Information Systems* **12**(2): 37–58.	Survey, Group Discussion, and Semi-Structured Interviews	Author explored the construct infrastructure flexibility and presented findings of the study, which measured the indicator of the construct.	Alignment of technological plans to business plans is presented as critical to infrastructure flexibility and efficacy. Study findings suggest that infrastructure flexibility might be affected by a kind of support from business, such as the need for infrastructure and IT leadership in planning for and managing certain resources. The author concludes that both business and IT capabilities may reflect the flexibility of the infrastructure components.
Dutta, S. (1996). Linking IT and Business Strategy: The role and responsibility of senior management, *European Management Journal* **14**(3): 255–269.	Case Study, $N=2$ (banks)	This study looked at the literature surrounding alignment and leadership. It also relied on the outsourcing literature.	This study looks at two cases that take two very different approaches to IT management. One company outsourced its IT while the other had it in house. Although the cases seem very different, various underlying issues are the same. There is a fundamental issue of literally managing IT, even if it is outsourced. Senior management must be involved in IT planning and implementation. In the cases studied, the outsourced IT was as successful as the internal IT group. The author provides some key take-aways for practitioners. First, managers can outsource IT, but cannot outsource the management of IT. Second, regardless of whether IT is in-house or outsourced, IT alignment will always be a big issue. Third, regardless of the structure of IT, active participation of management is vital. Fourth, age and lack of experience with IT have been found to be poor excuses for avoiding involvement in IT. Fifth, IT alignment will always be a problem if management simply views it as a function of the IT group (as opposed to a managerial responsibility). Finally, close interaction between the business and IT staff helps lay the necessary foundation for alignment.
Earl, M.J., Sampler, J.L. and Short, J.E. (1995). Strategies for Business Process Reengineering: Evidence	Case study $N=4$ UK organizations (3 large+1 mid-sized)	This paper reports results from case study research on relationships between business process	The paper proposes a framework for analysis based on the concept of alignment. This process alignment model comprises four lenses of inquiry: (1) process,

Citation (Author, year)	Research method	Theory/concept	Findings
from Field Studies, *Journal of Management Information Systems* **12**(1): 31–57.		reengineering (BPR), business strategy planning, and information systems (IS) planning.	(2) strategy, (3) information systems, and (4) change management and control. A taxonomy of strategies for BPR is derived from the four case studies: engineering, systems, bureaucratic, and ecological strategies. This taxonomy of polar types suggests a richer variety of BPR practice than has been documented to date and provides an opportunity and platform for further research.
Edwards, B.A. (2000). Chief Executive Officer Behaviour: The Catalyst for Strategic Alignment, *International Journal of Value-Based Management* **13**(1): 47–54.	Conceptual/Literature Review	This article relies on the literature on strategy (development, implementation, alignment), as well as the literature on the role of the CEO in the organization, and in particular, CEO behaviors vital to achieving alignment (Edwards, 1990).	This paper examines the CEO's role in strategic alignment, specifically as a catalyst to alignment. The author states that strategic alignment underpins strategic implementation. When implementing strategic change, the following behaviors are of the utmost importance: Communication, Strategic Interaction. Resource Utilization, and Leadership. More specifically, CEOs that engage in the following behaviors are thought to be the most effective: communicate, advise, interpret, disseminate information, act as spokesperson, lead, act as a role model, influence, provide personal support, show commitment, serve as liaison, interact with stakeholders, be involved, participate, marshal resources, allocate resources, build teams, coach, develop people, and clarify work. However, the author points out that although the CEO cannot and should not be solely responsible for alignment, they are the person in the organization that can act as the best catalyst for change.
Farrell, I.J. (2003). Aligning IT to Corporate Objectives: Organisational factors in use, Ph.D. thesis, Macquarie University, Sydney.	Case Study, $N=5$	The author outlines research looking at the impact of IT in organizations.	This study examined the factors that are used to achieve alignment in organizations. The author identified 21 factors that fell into three categories: management and planning, business, and technology. Management and planning includes CEO Attitude, CEO/CIO Relationship and Reporting Structure, CIO Management Style, Strategic Planning Quality, and IT Planning Methodology. Business includes Standards & Policies, Information Management, IT Services Management, Sourcing Methodology, Program Management, and Stakeholder Management. Technology includes IT Infrastructure Management, Internet Usage, Intranet Usage, Integrated Information

Citation (Author, year)	Research method	Theory/concept	Findings
			System, Data warehousing, Information Modeling, Reporting & Analysis Tools, E-Mail Usage, Standard Service Definitions, and SLAs. These factors exist in a hierarchy with Management & Planning providing the foundation for all other factors. CEO attitude was found to be the most important factor overall.
Feeny, D.F., Edwards, B.R. and Simpson, K.M. (1992). Understanding the CEO/CIO Relationship, *MIS Quarterly* **16**(4): 435–448.	Interviews, $N=12$ CEO–CIO pairs	This research relied on the literature of strategic information systems planning, the business/IS partnership, and CEO involvement in IT activities. This research also relied on the literature (especially the methods) regarding the study of relationships.	The goal of this research was to identify factors that led to a successful CEO/CIO relationship. A strong CEO–CIO relationship is thought to impact the organization in terms of its strategic information systems planning, business/IS partnerships, and CEO involvement in IT management. The authors found CEO, organizational, and CIO attributes that contributed to a strong relationship. The CEO attributes are a general management and/or marketing background, change-oriented leadership, attendance at IT educational seminars, experience with IT project success, perception of IT as critical to business, and perception of IT as agent to business transformation. Many of these factors relate to the CEO's attitude towards IS. Others are not so intuitive. For instance, the reason that a general management/marketing background was critical to success was that managers from those functional backgrounds tend to focus more on strategic issues. Attributes of successful CIOs are an analyst background or orientation, promotes IT as an agent of business transformation, contributions beyond the IT function, accurate views of CEO on IT issues, integration of IT issues with business planning, and a profile that stresses consultative leadership and creativity. Organizational attributes include a personal/informal executive style, executive workshops on strategic issues, and a CIO that is accepted as part of the executive team. Altogether, it is thought that those organizations that view IT as a vehicle with which to achieve transformation perform better than average.
Floyd, S.W. and Woolridge, B. (1990). Path Analysis of the Relationship Between Competitive Strategy, Information Technology, and	Longitudinal (3 years) case study $N=127$ questionnaires and 68 interviews (retail banks)	This research is based on that notion that technology has an effect on industrial organizations (Porter, 1980, 1985; Porter and Millar, 1985);	This study investigates competitive strategy's impact on IT, and IT's impact on organizational performance. Strategy appears to have direct effects on IT, and IT has direct effects on ROA. The study results reflect the

Citation (Author, year)	Research method	Theory/concept	Findings
Financial Performance, *Journal of Management Information Systems* 7(1): 47–64.		technology has become a strategic necessity (Clemons and Kimbrough, 1986); technology might provide competitive advantage (Arthur Anderson, 1985; Clemons and Row, 1988; Ives and Learmonth 1984; Johnston and Carrico, 1988; Johnston and Vitale, 1988; Wiseman, 1985); and that misalignment can decrease performance (Vitale *et al.*, 1986; Warner, 1987).	effect of competitive strategy on IT. However, the observed correlations seem to be industry specific. Strategic context makes a significant difference in the correlations observed between IT adoption and ROA. Based on the results, the authors imply that adoption of IT may sometimes be a necessary condition to successful strategy. The authors conclude that system implementation difficulties might explain misalignment. Successful strategy-IT alignment in some cases seems to be more complex than choosing a technology to enhance the effects from good strategy.
Galliers, R.D., Merali, Y. and Spearing, L. (1994). Coping with Information Technology? How British Executives Perceive the Key Information Systems Management Issues in the Mid-1990 s, *Journal of Information Technology* **9**(3): 223–238.	Survey $N=98$ senior executives	This paper is based on previous IS research and surveys assessing the views of managers. The aim of this paper is to provide comparisons of (i) the views expressed by IS executives *vis-á-vis* those with a non-IS role in UK organizations and (ii) the findings of this study with those of a previous similar British study undertaken in 1987.	Survey research among senior information systems (IS) and non-IS executives in UK organizations was conducted in order to identify their views as to the most important and problematic issues they face in managing information systems. The most critical IS management issues appear as one of the top five issues both in terms of importance and the problems currently being experienced: (1) improving IS strategic planning, (2) data as a corporate resource, (3) business process redesign, (4) developing an information architecture, and (5) quality of software development. The results of this survey help executives (both business and IT) to see more clearly how their views tally with colleagues in their own group and how in general their opinions differ from their colleagues in the other group. By discussing the apparent reasons for these differences of opinion, the authors hope that the culture gap between the two groups might diminish over time.
Galliers, R.D. (2004). Reflections on Information Systems Strategizing, in C. Avgerou, C. Ciborra and F. Land (eds.) *The Social Study of Information and Communication Technology*. (1st ed.,) London: Oxford University Press, pp. 231–262.	Conceptual paper	The author examines some of the key concepts and frameworks that have underpinned much of information systems strategy theory during this period, most notably, Earl's (1989) distinction between information systems and information technology strategies.	This chapter considers some of the more recent developments and new thinking in the field that have emerged over the last decade or so. It also acts with a view to pointing out future directions and current concerns, culminating in a proposed inclusive framework for information systems strategizing. By providing something of a historical account, an attempt has been made to draw together lessons from the past into the kind of cumulative account that has continued often to be missing from Information

Citation (Author, year)	Research method	Theory/concept	Findings
			Systems discourse (Keen, 1990). It is hoped that such reflection may prove useful to those interested in the social study of ICT, and not just those who share an interest in information systems strategy itself. Galliers attempts to summarize the previous research by developing an inclusive framework for information systems strategizing.
Goedvolk, H., van Schijndel, A., van Swede, V. and Tolido, R. (1997). Architecture, in D.B.B. Rijsenbrij (ed.) *The Design, Development and Deployment of ICT Systems in the 21st Century: Integrated Architecture Framework (IAF) ()*, http://home.hetnet.nl/~daanrijsenbrij/progx/eng/contents.htm: Cap Gemini Ernst and Young.	Conceptual	The authors review the development of information communication technologies (ICT).	This paper discusses the development of ProgrammeX, a framework by Cap Gemini Ernst and Young. ProgrammeX tries to answer the questions that managers may have regarding the adoption of new technology, faster application delivery, strategic alignment, transition of existing systems, reliability of complex distributed systems, and ICT as an enabler. The proposed solution includes web enterprise management, ICT enabled business transformation, integrated architecture framework and architectural design methods, component-based software development, and a delivery approach. The authors cite important ICT trends such as the global infrastructure, the communications power, the increasing reality of digital information, knowledge creation and exchanging, and the emergence of a virtual world.
Grover, V. and Sabherwal, R. (1989). An analysis of Research in Information Systems from the IS Executive's Perspective, *Information & Management* **16**(5): 233–246.	Quantitative literature review, $N=858$ published articles	This paper examines the extent to which research over the past decade has addressed those issues important to practitioners. A list of 26 key issues was used to categorize 858 articles published in four major IS outlets. The amount of research on each issue was then compared with the importance rating of the issue by practitioners over the decade 1977–1986.	Twenty-six issues of importance to IS practitioners fell into three distinct groups: (1) those for which research has apparently fallen short of the perceived importance, including the IS role and contribution and IS alignment in the organization; (2) issues on which there has been considerable research; possibly as a result of this effort, the issue has become less important, while research has continued unabated; (3) issues for which importance and publication percentages have both increased at a comparable pace. Study results suggest that IS research has fallen short of perceived performance in the case of global issues. Furthermore, the study indicates a disconcerting gap between what IS executives consider important and what is actually researched. It also demonstrates that IS researchers tend to concentrate on narrow technical issues rather than broad managerial issues. The

Citation (Author, year)	Research method	Theory/concept	Findings
			implications of this study strongly suggest that IS researchers should pay greater attention to the concerns of practicing managers.
Gupta, Y.P., Karimi, J. and Somers, T.M. (1997). Alignment of a Firm's Competitive Strategy and Information Technology Management Sophistication: The missing link, *IEEE Transactions on Engineering Management* **44**(4): 399–413.	Survey, $N=213$ IT executives from the financial services industry	IT management sophistication or IT maturity, which characterizes firms in terms of their evolution of IT. Competitive strategy as a match between the opportunities and risks inherent in the environment and firm's competencies. Competitive strategy and IT sophistication are empirically examined.	The study results confirmed the proposed instrument for measuring IT management sophistication. This instrument consists of four criteria: IT planning mode, IT control mode, IT organization and IT integration. The study shows that alignment between a firm's competitive strategy and IT management strategies can be translated into a particular set of distinctive competencies in IT management for each competitive strategy. Alignment requires top management to consider firm's competitive strategy and to limit their concerns to a few critical IT management processes. The findings of this study of financial services industry may not apply to less information-intensive industries, where IT plays more a supportive role.
Hanseth, O., Ciborra, C.U. and Braa, K. (2001). The Control Devolution: ERP and the Side Effects of Globalization, *The DATA BASE for Advances in Information Systems* **32**(4): 34–46.	Case study $N=1$ organization (Norsk Hydro)	This paper relies on the literature published on globalization, ERP and modernity. This research also looks at 'how to' implement ERP in organizations.	When looking at the implementation of ERP systems in large organizations, the typical business concerns are attaining the goals of the application, usually globalization and efficiency, securing the organization's acceptance, avoiding rigidity, and so on. This paper submits that there is a need to look at the broader context of ERP implementations. This paper argues that governance is a central issue in relation to ERP implementations, and a highly challenging one. Control and governance are core issues underlying virtually any strategy to develop and use IT solutions. The paper presents the case of the SAP implementation in Norsk Hydro, indicating how management perspective dictates the deployment agenda in every detail, but also how a number of positive and negative unexpected consequences of SAP can only be explained by the theory of modernity.
Hartung, S., Reich, B.H. and Benbasat, I. (2000). Information Technology Alignment in the Canadian Forces, *Canadian Journal of Administrative Sciences* **17**(4): 285–302.	Case Study, $N=8$ (Canadian Forces)	The authors based their research on the alignment perspective of Reich and Benbasat (2000) that looked at the social influence of alignment.	The authors examined levels of alignment in the Canadian Forces examining eight case sites. This research is important because is examined whether a model developed for the for-profit industry could be applied in the not-for-profit industry. The results indicate that shared domain of knowledge and

Citation (Author, year)	Research method	Theory/concept	Findings
			communications were positively related to alignment. However, the actions of a single leader also proved to be important. Many limitations of the model were found. Firstly, the Canadian forces had only recently begun to engage in long-term strategic planning and thus had a lack of maturity in their planning processes. Secondly, where there was low alignment, people could not articulate how IT would change the situation. Thirdly, the concept of shared domain of knowledge appeared to be very business-specific as it assumed a high degree of cross-functional experience and many interviewees in the military did not have enough experience at their current station to judge the situation. Overall, alignment in the military appears to be a function of shared domain of knowledge which leads to increased communication which has an impact on short-term alignment.
Henderson, J.C. and Venkatraman, N. (1992). Strategic Alignment: A Model for Organizational Transformation Through Information Technology, in T.A. Kocham and M. Useem (eds.) *Transforming organizations ()*. New York: Oxford University Press.	Conceptual	The authors develop and present the conceptual model of the potential influence of IT on competitive characteristics. The model links organizational transformation and the exploitation of IT capabilities in its competitive role.	Strategic alignment is a central element of organizational transformation. The proposed model is based on four key domains of strategic choice, namely (i) business strategy, (ii) organizational infrastructure and processes, (iii) IT strategy, and (iv) IT infrastructure and processes. The model has two distinct characteristics:(i) IT strategy is distinct from IT infrastructure and processes; (ii) the concept of strategic alignment is distinct from bivariate fit (i.e. linking only two domains) and cross-domain alignment (i.e. linking any three domains). The alignment model is descriptive (i.e. identifying four key factors to be considered), prescriptive (i.e. prescribing certain alternatives and approaches), and dynamic (i.e. requiring the organization to define approaches for transforming the internal organization structures and processes). Based on the discussion presented in the paper the author suggested that: (i) the effectiveness of strategic IT management will be significantly greater for any cross-domain perspective than any bivariate fit relationship, (ii) on average, the four cross-domain perspectives for strategic IT management will be equally effective, (iii) the effectiveness of strategic IT management is significantly greater for a complete

Citation (Author, year)	Research method	Theory/concept	Findings
			process than for any type of cross-domain alignment, under ceteris paribus condition, (iv) on average, the unidirectional perspective is the least effective process for strategic management, (v) on average, single-loop and focused processes will be equally effective and superior to unidirectional process for strategic IT management, (vi) on average, double-loop and focused processes will be equally effective and superior to unidirectional process for strategic IT management.
Henderson, J.C. and Sifonis, J.G. (1988). The Value of Strategic IS Planning: Understanding Consistency, Validity, and IS Markets, *MIS Quarterly* **12**(2): 187–200.	Case Study, $N=1$	The authors review the strategic planning process. The authors also discuss the internal and external consistency of this planning process. The need for cooperative behavior in the planning process is also addressed.	This article seeks to define effective strategic IS planning. The proposed planning process consists of three phases: (1) business strategy formulation; (2) IS strategic formulation; and (3) action plan and resource allocation. Business strategy formulation is a vital first step as IS strategic formulation assumes a clear vision and strategy for the organization. IS strategic formulation uses the goals established in the business strategy formulation to provide a direct linkage to the IS plan. In order to do this, the authors recommend that a critical success factor approach be employed. The critical success factors process sets the foundation for developing critical assumption sets, critical decision sets, value-based processes, and strategic data models. Of critical importance is the strategic fit both internally and externally. In the case study presented, the company followed the proposed strategic IS planning process. However, the authors note the iterative nature of the process (not to be confused with incremental). When a strategic plan is developed, it should be 'tested' and if need be, changed.
Henderson, J.C. and Venkatraman, N. (1993). Strategic Alignment: Leveraging Information Technology for Transforming Organizations, *IBM Systems Journal* **32**(1): 4–16.	Conceptual	The authors present the strategic alignment model for conceptualizing and directing strategic management of IT (Henderson and Venkatraman, 1992, 1989).	The authors suggest that managers are more comfortable with their capability to comprehend business positioning choices rather than IT positioning choices, i.e. strategic decisions to obtain critical technology to support business strategies. The authors also articulate the need to align internal and external domains of IT. Once aligned, these domains should be integrated with business domains. The proposed model suggests that effective management of IT requires balance between the four domains of the model, where business strategy is a driving force and

Citation (Author, year)	Research method	Theory/concept	Findings
			IT strategy is an enabler. Finally, the authors underline the intrinsic nature of the model, which requires four domains of the model to be changed, depending on shifts in the business environment.
Henderson, J.C., Venkatraman, N. and Oldach, S. (1996). Aligning Business and IT Strategies, in J.N. Luftman (ed.) *Competing in the Information Age – Strategic Alignment in Practice*. Oxford: Oxford University Press.	Non-empirical study	This chapter summarizes the most recent evolution of the strategic alignment model developed by Henderson and Venkatraman (1989, 1992, 1993), and the four key management mechanisms to assist both short-term decision taking and long-term strategic planning (value management, governance, technological and organizational capabilities).	The alignment perspective is expanded to emphasize that the redesign of business processes can better support the business strategy with the strategy formulator role of top executives and strategy implementer role of IS management. The IT strategy should be developed as a response to a business strategy. However, a business strategy can be modified via emerging IT capabilities. The implications of strategic fit are reflected in terms of the products and services provided to the organization. The four identified key alignment mechanisms help organizations achieve strategic alignment and create the dynamic capabilities to transform the corporation effectively. The alignment, however, has a dynamic nature, which requires continual assessment of trends across these four domains, evolving from one perspective to another based on the shifts in the business environments, both internal and external. The authors identify the lack of understanding of the enabling strategic choices as a major reason for the dissatisfaction with the level of integration.
Hirschheim, R. and Sabherwal, R. (2001). Detours in the Path Toward Strategic Information Systems Alignment, *California Management Review* **44**(1): 87–108.	Case study $N=3$ organizations	This study uses an adaptation of the strategic orientation typology of Miles and Snow (1978) and adopts a multi-dimensional view of strategic IS from Henderson and Venkatraman (1993).	This study identifies three alignment profiles: infusion (alignment through business leadership), alliance (alignment through partnership), and utility (alignment through low-cost delivery). Together, the three strategic IS alignment profiles, the explicit recognition of the possible problematic trajectories, and the identification of key factors explaining these trajectories provide a framework organizations should find useful in pursuing their strategic IS realignment efforts. The key lesson from three case studies is that it is important for organizations to understand the dynamic and emergent nature of business-information systems alignment.
Ho, C. (1996). Information Technology Implementation Strategies for Manufacturing	Conceptual	This article reviews the literature on strategic IT, manufacturing strategy, and the strategic alignment model	The authors develop a model of bivariate fit and cross-domain alignment that incorporates both elements of IT strategy and manufacturing strategy.

Citation (Author, year)	Research method	Theory/concept	Findings
Organizations: A Strategic Alignment Approach, *International Journal of Operations & Production Management* **16**(7): 77–100.		(Henderson and Venkatraman, 1993). The authors use the SAM to build their evolutionary process.	Given that manufacturing is vital to business strategy and that manufacturing is highly information-intensive, achieving alignment between an organization's business, IT, and manufacturing domains is important. The author's proposed model is achieved via four distinct stages: (1) technology implementation; (2) technology exploitation; (3) strategy implementation; and (4) strategy sustenance. In stage 1, management views manufacturing as neutral and simply seeks to minimize any negative effects from it. At stage 2, management seeks to achieve parity with competitors and be externally neutral. During stage 3, management uses manufacturing to provide credible support to business strategy. During stage 4, a stage that is reached only by world-class companies, management pursues a manufacturing-based competitive strategy that is externally supportive. The author distinguish their model from a just in time (JIT) model and a materials requirement planning (MRP) model. The proposed alignment model underscores these models as efficient use of manufacturing technology is a pre-condition for success in both. The author implies that practitioners need to try to evolve their use of manufacturing technology so it becomes more of a strategic weapon.
Huff, S.L. and Chan, Y E. (1999). Aligning Information Systems Strategy with Business Strategy, in M. Fleck (ed.) *Managing for Success,* 1st edn., Toronto: HarperCollins Publishers, pp. 43–47.	Practitioner Guide	The authors discuss Venkatraman's seven characterizations of strategic orientations: aggressiveness, analysis, defensiveness, futurity, innovativeness, proactiveness and riskiness.	This chapter discusses previous research conducted by the authors. They break the research down into practical steps that managers can take in order to achieve alignment. Strategic alignment is positively related to organizational performance and IS effectiveness; thus, managers should invest sparingly in alignment. Managers should devote resources to all seven areas of strategy in order to minimally comply with all. Managers should first devote resources to IS planning, then to action, and then to analysis in order to reap the most benefits. Organizations that do not devote attention and resources to alignment may miss out on important opportunities.
Hunt, G.E. (1993). Management Challenges for Survival in the 1990 s,	Literature Review	This article does extensive alignment literature review and draws results from various surveys and other	This paper presents the argument that the key to corporate success is related to the effective management of information technology (IT). In many

Citation (Author, year)	Research method	Theory/concept	Findings
Journal of Information Technology **8**(1): 43–49.		research in US and UK. (The material described in this paper formed the basis of the 1992 AIT lecture at the City University Business School.)	cases, this will require a new style of management with vision and commitment from the CEO. Top management must aspire to what can be done through IT, using their understanding of the particular business issues in their company and their imagination when conceiving business strategies. For companies to succeed in the 1990 s and beyond, they must face the challenge of the 1990 s – the effective marriage between business and IT. But above all, top management must be in charge, and that means being in charge of IT too.
ICEX, 1999	Literature Review	The paper presents the construct IT strategic alignment based on earlier studies in the filed (Broadbent, 1998; Henderson and Venkatraman, 1993; Chan, 1999) and characteristics of effective IT strategic alignment.	This paper is aimed more at the practitioner. The authors cite four critical factors for CIO, which are highly related to strategic alignment, namely building shared vision, building relationships, CEO relationship and proactive planning. IT strategic alignment can be multidimensional and dynamic. Alignment, however, is a complex and multifaceted process (a journey) that can never be completely achieved.
Irani, Z. (2002). Information Systems Evaluation: Navigating Through the Problem Domain, *Information Management* **40**(1): 11–24.	Case study, $N = 1$	This paper reviews the normative literature outlining the problems associated with IS evaluation.	This article presents a framework for evaluating IS investments. The case in the article looks specifically at manufacturing resource planning (MRP). When making investment decisions, this paper argues that three core areas should be focused on: the justification processes (concept and financial bases); the limitations inherent in traditional appraisal techniques; and life cycle evaluation. Concept justification looks at the softer side of investment decisions. It takes into account the views of strategic and operational stakeholders, recognizing that IT investment takes place over a long period of time and in the larger organization (thus, impacting a greater number of individuals). The author proposes that increased concept justification will lead to increased support for the investment. Financial justification refers to the more traditional appraisal methods of projects. This justification is performed by both the management and by the accountants. However, there are many limitations with this procedure including the unquantifiable nature of certain IT projects and the lack of long-term perspectives taken by financial analyses. Life cycle evaluation calls for a post-implementation

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			review of IT investments. The author proposes that the scope of the benefits and costs used for justifying IT investments be contingent on the already available indicators in an organization. The author also proposes that there is a need to classify the benefits into strategic, tactical, and operational, and further classify each sub-category into financial, non-financial and intangible. In the case study, the propositions were tested. The authors found that specific evaluation criteria made the project investment easier (financial justification). The author also found that human impact and involvement in the project was vital, and human support was necessary for project success (concept justification). Evidence of spiraling out of control indirect costs was found. Hence, there was no relationship between the measurement of a specific benefit and the use of those benefits during the investment justification. However, many benefits and costs were found *post hoc*.
Jarvanpaa, S. and Ives, B. (1993). Organizing for Global Competition: The fit of Information Technology, *Decision Sciences* **24**(3): 547–580.	Survey, $N=109$, followed by a second questionnaire sent to the first 70 respondents, $N=53$	This paper utilizes organizational information processing theory which states that organizational structures required to deal with information processing requirements are derived from a firm's environment and tasks. This paper also utilizes organizational contingency theory. This paper also utilizes Bartlett and Ghoshal's notion of global strategic orientation, namely national responsiveness, efficiency, and shared learning; and their classification of international firms, as global, transnational, multinational or international. Finally, the authors utilize the theoretical notions of fit.	This paper examines the fit between IT and global business decision-making structures without measuring performance. The authors state that studying global business is important since it faces increased challenges coordinating units and dealing with the dynamic nature of the international market. However, the authors also feel that global organizations have more resources and much to gain by investing in IT alignment. Overall, the authors predict global firms will attempt to match IT with what they perceive to be the business strategy of the firm. The authors form their hypotheses around the view of IT in the organization (independent operations, headquarters driven, intellectual synergy, and integrated IT) against various IT dimensions (locus of IT decision making, common systems, mode of operations, IT reporting relationship, and development approach). Hypothesis 1 predicts that globally competing firms that report to follow different IT configurations exhibit different and distinct ways of structuring IT activities. Hypothesis 2 predicts that the firm's management designs the IT configuration to be

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			consistent with the perceptions of the organizational structure. Hypothesis 1 was generally supported, with different IT dimensions being implemented differently depending on the view of IT in the organization. However, hypothesis 2 only received weak support, with almost half of the firms in the sample having a misfit. Potential reasons for this misfit were explored in follow-up interviews. When misfit was caused by centralization, it was attributed to cost pressures aimed at decreasing the IT budget worldwide and/or the belief that it was necessary to build a consistent architecture. When misfit was observed in cases where there was high business integration and low IT integration it was attributed to a lack of managerial focus on IT issues and/or cost pressures. The authors recommend many areas for future research. They suggest future research take a more holistic approach to fit. They also suggest that future research attempt to view alignment as more of a process. They also suggest that research be done to see how technology drives a firm's strategy. More generally, the authors suggest that future research try to incorporate a more global perspective.
Jarvenpaa, S.L. and Ives, B. (1990). Information Technology and Corporate Strategy: A view from the top, *Information Systems Research* **1**(4): 351–376.	Longitudinal case study (8 years, data drawn from 649 annual reports to shareholders of 88 companies)	This paper looks at the content of CEO letter as being a good proxy for strategic priorities. This research also extends past research on IT support from organizational leadership.	In this study the authors develop the following premises: • To exploit strategic opportunities from IT, CEOs must view IT as s component of corporate strategy. • The strategic role of IT and the CEO's perceptions of the importance of IT vary across industries. • Information technologies' hypothesized strategic role is frequently postulated to have grown over time. Industry leaders make more aggressive use of IT than 'losers'. The study also reveals that CEOs from different industries perceive the role of IT differently. The study makes a modest attempt to examine the alternative explanations for the profit/IT investment relationship and finds evidence for a reciprocal relationship. The role of IT within a firm has been repositioned over time and the extent of changes varies across different industries. The research

Citation (Author, year)	Research method	Theory/concept	Findings
			(tentatively) suggests that it is no longer worthwhile to discuss the generic notion of information for competitive advantage.
Jarvenpaa, S.L. and Ives, B. (1991). Executive Involvement and Participation in the Management of Information Technology, *MIS Quarterly* **15**(2): 205–227.	Case study (data drawn from annual letters to shareholders, industry handbooks and follow-up interviews)	The concept of executive support or executive involvement, which refers to the CEOs' activities or substantive personal interventions in the management of IT, is explored. However, participation and involvement are viewed as two distinct constructs (Baraki and Hartwick, 1989).	The study reveals that executive involvement, executive participation, executive age and functional background are significantly associated with the progressive use of IT. The findings suggest that executive involvement is more strongly associated with the firm's progressive use of IT than executive participation in IT activities.
Johnston, H.R. and Carrico, S.R. (1988). Developing Capabilities to use Information Strategically, *MIS Quarterly* **12**(1): 37–48.	Field study, $N=90$ interviews across 18 corporations in 11 industries	This research is based on the notion that IT plays a crucial role in business strategy, but exactly how an advantage is gained is unknown.	This article looks at the development and use of internal capabilities. The authors found that competitive advantage depends on the interaction between industry conditions and internal capabilities to identify and exploit opportunities. The aim of this study was to help manager ascertain the extent to which the firm's competitive environment contains opportunities to use technology strategically, evaluate an organization's readiness to undertake an IT strategy, and to develop internal IT capabilities. There are three major findings of the study: (1) industry and environmental factors influence the direction and pace of strategic deployment of information technology; (2) there is significant variation in the degree to which IT is integrated into primary strategy or specific business units; and (3) several conditions are consistently present in those organizations who strategically utilize IT. Various environmental factors lead to the strategic deployment of IT capabilities including significant content in key business relationships, firms that specialize in products or services that have a limited or no life span (i.e., airplane seats), and increased competitive pressure within the industry. The authors were also able to classify organizations with regard to how integrated IT was in the overall organization. This classification was 'Traditional', where IT supports operations but is not related to business strategy; 'Evolving', where IT supports business strategy; and

Citation (Author, year)	Research method	Theory/concept	Findings
			'Integrated', where IT is integral to strategy. The authors also identified five internal factors that helped IT support strategy, namely, leadership, the integration of the IS function and the strategy function in the organization, direct contact between the IT function and the line business, capability in the IS function, and the mechanisms available for the line function to influence IT.
Johnston, K.D. and Yetton, P.W. (1996). Integrating Information Technology Divisions in a Bank Merger: Fit, Compatibility and Models of Change, *Journal of Strategic Information Systems* **5**(3): 189–211.	Case study	The authors create a framework using Mintzberg's (1979) five organizational configurations and Scott Morton's (1991) MIT model.	The authors developed a framework to analyze the integration of two IT divisions in a bank merger. They present the normative implications for managing IT integration and examine how issues of fit and compatibility constrain the choice of a merger strategy and model of integration. The authors identify a set of ideal type configurations of contextual and organizational elements that maximize fit. The study reveals that the compatibility of IT configurations influences the effectiveness of different models of change for IT integration. When individual organizational components are compared, each bank's IT configuration resembles a different ideal organizational type defined by Mintzberg, namely a machine bureaucracy and a divisional form. Based on the results, the authors identify three models of change in IT integration, namely (i) best of breed, (ii) absorption and (iii) co-existence. The short-term need to reduce the complexity and risk created by incompatibility is a key driver of IT integration.
Kaarst-Brown, M.L. and Robey, D. (1999). More on Myth, Magic and Metaphor: Cultural Insights into the Management of Information Technology in Organizations, *Information Technology & People* **12**(2): 192–218.	Ethnographic case study, $N=2$	This article reviews the literature looking at organizational culture's impact on IT and the use of metaphors in helping to explain complex organizational problems. This paper also utilizes grounded theory.	This research investigates the cultural assumptions regarding IT and IT management. Citing previous research that alludes to IT as 'magic', the authors use the metaphor of the magic dragon and apply it to IT in the organizational context. The authors develop five cultural archetypes. The first, the magic dragon sitting on a pile of gold, is a culture in which IT is revered. This creates a very positive environment for IT to develop and prospers, but can ironically lead to complacency if the company places all trust and assumes that IT is being utilized effectively. The second archetype, the caged dragon, is one in which the IT group is extremely confined and limited in their scope. This ensures that IT is subordinated to business

Citation (Author, year)	Research method	Theory/concept	Findings
			and IT specialists do not develop the necessary skills to add to the organization. The third archetype, pet dragons, makes reference to the sorcerer's apprentice fear that a little bit of knowledge is a dangerous thing. This archetype argues for decentralized IT, but lacks the power to develop a strong and unified IT model. The fourth archetype, team dragons, represents an integrated model of IT. This model can be very beneficial, but oftentimes the 'team process' turns into a ritual as opposed to an actual team process. This negates any benefits. The fifth and final archetype, dead dragons, represents a culture fearful of IT. An obsessive fear of IT can stunt innovation and lead to the under-utilization of systems.
Kanellis, P., Lycett, M. and Paul, R.J. (1999). Evaluating Business Information Systems Fit: From concept to practical application, *European Journal of Information Systems* **8**(1): 65–76.	Case study based on survey	This paper applies Repertory Grids (Kelly, 1955). The authors present the framework to measure fit along three processes: (i) decision making, (ii) innovation and information acquisition and (iii) distribution.	The authors concluded that poor fit in this case results from the inability of the information system to respond to changes in the business case. This study supports the view that at any time, an information system can only approximate the changing requirements emerging from the constantly shifting trends and policies of organizational life (Hirschheim *et al*., 1995). The authors suggest that improvement in intra-unit integration could be achieved when the information systems allows for sharing a quick and simple work flow between a small number of individuals. This paper presents an interactive approach to the measurement of information systems success, and serves as a pluralistic approach to interpret case studies. It delivers both researchers and practitioners an '*in situ*' understanding of the information system that can serve as the basis for corrective action.
Kearns, G.S. and Lederer, A.L. (2000). The Effect of Strategic Alignment on the use of IS-Based Resources for Competitive Advantage, *Journal of Strategic Information Systems* **9**(4): 265–293.	Survey, $N=107$ matched pairs of CIO and CEO	The research model anticipates the existence of two alignment models: • ISP-BP alignment of the IS plan with the business plan, which signifies IS management's understanding of business strategy (Reich and Benbasat, 1996). • BP-ISP alignment of the business	The study proves the dichotomy of alignment. Alignment of the IS plan with the business plan is associated with the use of IS for competitive advantage by IS executives as well as other senior executives. Alignment of the business plan with the IS plan is associated with the use of IS for competitive advantage by senior IS executives, but not by other senior executives. The study results support the expectation that alignment of IS with business plans produces the

Citation (Author, year)	Research method	Theory/concept	Findings
		plan with the IS plan, which ensures that the business plan reflects the experience and knowledge of the organization utilizing IS-based resources, and signifies better top management understanding and commitment (Bensaou and Earl, 1998).	use of IS-based resources for competitive advantage. The findings also support the expectation that the IS executive perspective alignment of business with IS plans produces the use of IS-based resources for competitive advantage. However, the expectations that from the business executive perspective alignment of business with IS plan could produce competitive advantage is not supported. The authors point out that no single reason irrefutably explains the failure to support the relationship between BP-ISP alignment and the use of information-based resources for competitive advantage with data from business executives. However, this failure, according to the authors, is very noteworthy.
Kearns, G.S. and Lederer, A.L. (2003). A Resource-Based View of Strategic IT Alignment: How Knowledge Sharing Creates Competitive Advantage, *Decision Sciences* **34**(1): 1–29.	Survey, $N=161$ CIOs	This paper uses the resource-based view of the firm that posits that a firm's internal assets and processes are most vital for competitive advantage.	This research investigates the importance of the formal planning approach taken by CEOs and CIOs in IT planning. The authors develop and test a model. This model predicts that the information intensity of an organization will lead greater involvement of the CEO and the CIO in the IT and business planning processes. This in turn will lead to the business and IT plans reflecting each other, which is hypothesized to lead to IT that is used for competitive advantage. All paths are supported with the exception of an IT plan that reflects the business plan leading to more strategic use of IT. This study has many key findings. Firstly, information intensity plays a significant role in involvement in planning processes that leads to an increase in knowledge sharing. However, some interesting findings emerged. The CEO's participation in IT planning did not clearly lead to the IT plan reflecting the business plan, but. This can be attributed to the fact that the CIO does not perceive their participation as important and/or the CEO's involvement may only be symbolic. These finding present a fruitful area for future research.
Kim, K.K. and Michelman, J E. (1990). An Examination of Factors for the Strategic use of Information Systems in the Healthcare Industry, *MIS Quarterly* **14**(2): 201–215.	Conceptual using case studies to reinforce propositions ($N=4$). Case studies were not elaborated on in this	This article adapts the work of Porter on generic competitive strategy that outlines three strategic types: overall cost leadership, product differentiation, and special market	This article looks at the strategic use of information systems technology (IST) in the healthcare industry. The authors develop a model with three factors. The first factor involves over-coming political issues in order to achieve integration such as the barriers

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
	article nor were methods discussed.	focus. This research also utilizes Porter's Five Forces model. The authors also review the literature on the benefits of IST.	between administrators and physicians, where physicians act as gatekeepers. This leads to the first proposition: for the integration of health information systems (HIS) to be successful, the conflict between physicians and administrators is an important factor that a hospital must effectively deal with. The second factor in the model involves the integration of the transaction processing systems (TPS) and the information reporting systems (IRS). This leads to the author's second proposition: Integration of isolated TPS and IRS systems throughout the organization is an important factor in the strategic use of healthcare information systems. The third factor in the model relates to the strategic use of the integrated TPS/IRS, leading to the third proposition: healthcare organizations can achieve competitive advantage through the successful application of TPS/IRS to strategic areas. The authors also address how these propositions can be applied in the healthcare industry by developing a healthcare-specific five-force model. The five forces in the healthcare industry are patients, suppliers, substitutes, third-party payers, and inter-industry competition. The authors suggest that future research should empirically test the propositions developed in this article. Secondly, future research should also examine the link between information and strategy development and organizational adaptation in the healthcare industry. Thirdly, future research should examine the political and conflict issues surrounding alignment. The authors suggest that practitioners should pay particular attention to the political environment surrounding IST, focus on integrating the current isolated systems, and try to increase user participation on the systems.
Lederer, A.L. and Mendelow, A.L. (1989). Coordination of Information Systems Plans with Business Plans, *Journal of Management Information Systems* **6**(2): 5–19.	Case study $N=20$ organizations	Coordination of the IS plan and the business plan can be achieved in three dimensions (Shank *et al*., 1973): Content, Timing, and Personnel.	The coordination of IS plans with business plans can be an essential element enabling and organization to realize its objectives. However, earlier research suggested that IS executives sometimes encounter difficulty achieving this coordination. The research described in this article uncovered four general reasons for this difficulty, and four general actions that

Citation (Author, year)	Research method	Theory/concept	Findings
			IS executives take to reduce the difficulty. The four reasons for this difficulty include, unclear or unstable business mission, objectives and priorities; lack of communication; absence of IS management from business planning process; and unrealistic and lack of sophistication of user managers. The actions that IS managers can take to reduce these issues include, encourage business management's participation in IS planning; establish an IS plan; rely on business management's planning process; and participate in business management's planning process. Moreover, the research suggested that the actions of IS executives are successful only if the coordination is mandated by top management.
Lee, S. and Leifer, R.P. (1992). A Framework for Linking the Structure of Information Systems with Organizational Requirements for Information Sharing, *Journal of Management Information Systems* **8**(4): 27–45.	Conceptual	The authors review the general structural relationships between organizations and IS. Then, they define IS structures and present a model for determinants of IS structure. Following that, based on the construct of information sharing and contingencies for information sharing, the authors develop a series of propositions, which link emerging organizational structures with emerging IS structures. Later, the authors review the traditional organizational structures and finally, they discuss managerial implications and conclusions.	Based on the construct of information sharing and contingencies for information sharing, the authors presented 11 propositions linking emerging organizational structures with emerging IS structures. The authors imply that information sharing is a key concept for linking organizational structures with IS structures. They also suggest that proper alignment between organizational and IS structures is critical for organizations to achieve flexibility and efficiency in competitive and turbulent environments. The study indicates that high levels of information sharing capabilities can be achieved by recentralizing IS and systems integration in the form of consolidated databases, cross-functional integration, and in the development of inter-organizational network systems.
Leiblein, M.J., Reuer, J.J. and Dalsace, F. (2002). Do Make or Buy Decision Matter? The Influence of Organizational Governance on Technological Performance, *Strategic Management Journal* **23**(9): 817–833.	Case study (based on a report of firm-level activity of 176 global circuit manufacturers)	The authors utilize the literature on outsourcing and vertical integration as factors enhancing performance. The authors use transaction cost economic theory (TCE) and its fundamental principle that efficiency will be enhanced when a fit exists between the chosen governance arrangement and the underling attributes of that transaction and the	The study present two main findings. The first finding is that contrary to the popular belief that vertical integration or outsourcing leads to superior technological performance, the results in this study indicate that governance decisions do not significantly influence technological performance directly, but rather performance is driven by factors underlying governance choice. Secondly, deviation from the optimal form of governance may have a negative impact on performance; especially technological performance might be negatively influenced by

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
		broader contracting environment (Williamson, 1985, 1991).	inadequate contractual precautions. The authors also indicate that future research should reveal other factors influencing governance choice, which in turn influence technological performance. This is in line with the premise that the optimal form of governance is likely to be contingent on both attributes of the transaction and the pre-existing strengths and weaknesses of the focal firm (Wiliamson, 1998, 1999).
Levy, M., Powell, P. and Yetton, P. (2001). SMEs: Aligning IS and the Strategic Context, *Journal of Information Technology* **16**(1): 133–144.	Longitudinal case study (based on 27 cases studies of SMEs undertaken between 1995 and 1999)	The authors develop the focus-dominance model, which proposes that an SME's strategic context influences investment in IS. The authors present four categories of alignment of business and IS for SMEs in each of the four quadrants of the focus-dominance model. Quadrants of the model correspond to customer focus (high, low) and strategic focus (high, low). The authors assume that IS investment in SMEs is successful when it is either a low-cost investment for providing efficiency savings or the enabling of value added strategy.	The authors develop the focus-dominance model and validate that SMEs align there IS with their strategic context as defined by the model. Since there is a significant difference between the IS profiles for the SME in each of the four quadrants in the focus-dominance model, the results strongly support the hypothesis that the adoption of IS by SMEs is a function of the strategic context – as defined by the focus-dominance model. The authors suggest that tactical Information Systems (e.g., e-mail) do not need to be aligned to the strategic context as opposed to the strategic IS, which need to be aligned. The authors imply that business dynamism is also a function of age and experience, which proves the previous theories that age and experience of the owner is frequently the most important factor in IS-based success (Palvia and Palvia, 1999). The findings of this study are consistent with literature in this area of focused and contingent IS investment by SMEs, such as Cragg and King (1992). However, in contrast with the above mentioned researchers, the study shows that managers of SMEs do align their IS and strategic contexts with the expectation of collecting significant benefits.
Lockamy, A.I. and Smith, W.I. (1997). A Strategic Alignment Approach for Effective Business Process Reengineering: Linking Strategy, Process and Customers for Competitive Advantage, *International Journal of Production Economies* **50**(2–3): 141–153.	Case Study, $N = 1$	The authors review the literature on business process reengineering.	This article provides a conceptual framework and key propositions that aid in more successful business process reengineering (BPR). The authors outline three principles for strategic alignment: processes selected for reengineering must have a strategic impact on the firm; processes selected for reengineering must have a significant impact on customer satisfaction and delight; and the strategic direction of the firm must be driven by customer requirements for effective business

Citation (Author, year)	Research method	Theory/concept	Findings
			process reengineering. The authors outline three principles for performance measurement: performance measures are needed to evaluate the potential impact of a reengineering program on competitiveness and customer satisfaction; process measures are needed to assess the current performance and reveal future improvement opportunities; and the measurement system must provide a mechanism for determining if the current strategies and processes are sufficient relative to a particular customer base. The authors provide three information technology principles: information technology must be used to facilitate easy access to process information across function barriers; the assessment of information technology for use in a BPR program must be conducted within the context of the firm's information technology strategy; and the deployment of information technology for use in a business process reengineering program should not occur until the selected processes have been successfully redesigned. The authors also outline three principles for reengineering methodology: BPR programs should adopt a pilot implementation approach to minimize risk; BPR programs should utilize a project management approach to control cost, performance, and timeliness objectives; and BPR programs must evaluate alternative approaches to process redesign to promote creativity and innovation.
Luftman, J.N. (1996). Applying the Strategic Alignment Model, in J.N. Luftman (ed.) *Competing in the Information Age,* 1st edn., New York: Oxford University Press, pp. 43–69.	Conceptual	This article applies the strategic alignment model (Henderson and Venkatraman, 1993).	This article details how the SAM is utilized. The author uses the concept of cycling through the four domains of the model: business strategy, IT strategy, IT infrastructure, and organizational infrastructure. The starting point of the cycle is called the 'domain anchor'. The cycle then pivots through a second adjoining box called the 'domain pivot'. Finally, the third box is called the 'domain impact'. The author identifies the four most prevalent ways to cycle through the SAM: strategy execution perspective, technology potential perspective, competitive potential perspective and the service level perspective. The strategy execution perspective is anchored in business strategy, pivots in organizational infrastructure, and impacts IT infrastructure. This

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			perspective sees the role of executives as being the definers and communicators of strategy (top-down approach). IT managers must then implement business strategy like any other functional manager. IT is measured using a financial or productivity measure of IT value. The appropriate strategic planning approach is either business process reengineering or IT planning. The technology potential perspective is anchored in business strategy, pivots in IT strategy, and impacts IT infrastructure. This perspective views executives as the technological visionaries and then IT managers as IT architects. IT is measured by its added value to performance. The appropriate strategy planning approach is to devise an IT strategy. The competitive potential perspective is anchored in IT strategy, pivots in business strategy, and impacts organizational infrastructure. This perspective sees the top executive team as being a business visionary and IT management play the role of business catalyst and architect. IT is valued based on the increased competitive potential gained from its implementation. The appropriate competitive planning method for competitive potential is to leverage the IT strategy into a business strategy so it reflects the updated technology. The service level perspective is anchored in IT strategy, pivots in IT infrastructure, and impacts organizational infrastructure. This perspective requires executives to prioritize IT projects while IT managers are the service providers. IT is evaluated on its end-user value. IT should be planned based on service level perspective, either IT planning, IT process reengineering, or executing a previously defined project. The author then quickly mentions four other less common cycles: organizational infrastructure perspective, IT infrastructure perspective, IT organizational infrastructure, and Organizational infrastructure strategy. The concept of fusion – or when two cycles are performed simultaneously – is discussed. Four types of fusion are discussed. First, IT infrastructure fusion is a combination of the strategy execution perspective and the technology potential

Citation (Author, year)	Research method	Theory/concept	Findings
			perspective. The organizational infrastructure fusion is a combination of the competitive potential perspective and the service level perspective. The business strategy fusion is a combination of the IT infrastructure strategy and IT organizational infrastructure. Finally, IT strategy fusion is a combination of organizational IT infrastructure and organizational infrastructure strategy.
Luftman, J., Kempaiah, R. and Nash, E. (2006). Key Issues for IT Executives 2005, *MIS Quarterly Executive* **5**(2): 81–101.	Survey, $N=105$	Not applicable.	This article reports the results of a formal survey of managers about their top IS concerns. Specifically, this survey uncovers opinions on four topics: key management concerns, application and technology development, organizational consideration and the enablers and inhibitors of IT and business alignment. Overall, the top five management concerns in 2005 were: (1) IT and business alignment; (2) attracting, developing and retaining IT professionals; (3) security and privacy; (4) IT strategic planning; and (5) business process reengineering. Alignment was the number one concern across all industries and alignment has been the number one issue of concern since 2003. However, another fast mover on the list was security technology. Thus, alignment and security issues are perceived to be of greatest importance to IT professional.
Luftman, J. (2000). Assessing Business-IT Alignment Maturity, *Communications of the Association for Information Systems*, 4, Article 14.	Primarily conceptual, although some case evidence is cited (no methods)	This paper reviews the literature outlining why strategic alignment is important and why the maturity level in this process is important.	This paper develops a method to evaluate the strategic alignment maturity of an organization. Mature alignment is achieved when business and other functions (i.e., HR, marketing, etc.) adapt their strategies with IT. The authors propose five levels of strategic alignment maturity: (1) Initial/Ad Hoc Process; (2) Committed Process; (3) Established Focused Process; (4) Improved/Managed Process; and (5) Optimized Process. Each stage is evaluated based on the following criteria: communications maturity, competency/value measurement maturity, governance maturity, partnership maturity, scope and architecture maturity, and skills maturity. Each of these criteria is assessed individually and then an overall level of maturity is given. It is important that this assessment is done in a team of both business and IT executives. The goal of the maturity assessment is to provide the

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			organization with areas of improvement. Most organizations are at level 2 of strategic alignment maturity.
Luftman, J.A., Papp, R. and Brier, T. (1999). Enablers and Inhibitors of Business-IT Alignment, *Communications of the Association for Information Systems*, 1, Article 11.	Longitudinal study $N=500$+firms and responses from 1051 executives	The strategic alignment model, suggested by Henderson and Venkatraman (1989, 1993, 1996), was applied by the authors throughout this research.	The authors found that there are known enablers and inhibitors that help and hinder alignment. What is striking about these are that the same set of topics (executive support, understanding the business, IT-business relations, and leadership) show up as both enablers and inhibitors. The authors conclude that strategic alignment is an ongoing process. Executives should work toward minimizing those activities that inhibit alignment and maximizing those activities that bolster it, such as improving the relationships between the business and IT functional areas, working toward mutual cooperation and participation in strategy development, maintaining executive support, and prioritizing projects more effectively. In previous work (Luftman, Papp and Brier, 1999), the authors have presented the detailed findings of the enablers-inhibitors study. The purpose of this article is to present the methodology that they have applied that leverages the enablers and inhibitors.
Luftman, J. and Brier, T. (1999). Achieving and Sustaining Business-IT Alignment, *California Management Review* **42**(1): 109–122	Longitudinal study $N=500$+firms and responses from 1051 executives	The strategic alignment model, suggested by Henderson and Venkatraman (1989, 1993, 1996), was applied by the authors throughout this research.	Achieving alignment, as an evolutionary and dynamic process, demands focusing on maximizing the enablers and minimizing inhibitors; it requires strong support from senior management, good working relationships, strong leadership, appropriate prioritization, trust and effective communication. The six most frequent enablers of alignment are: senior non-IT executives' support for IT, IT involvement in strategy development, IT understanding of the business, business–IT partnership, well-prioritized projects, IT demonstration of leadership. The six most frequent inhibitors of alignment are: lack of close relationship between IT and business, IT does not prioritize well, failure of IT to meet its commitments, IT lack of understanding of business, lack of senior executives' support of IT, IT management lacks leadership.

Citation (Author, year)	Research method	Theory/concept	Findings
Luftman, J.N., Lewis, P.R. and Oldach, S.H. (1993). Transforming the Enterprise: The Alignment of Business and Information Technology Strategies, *IBM Systems Journal* **32**(1): 198–221.	Conceptual, applied	The article reviews the literature on the need for business transformation and IT's role in transformation. The authors base their research on the strategic alignment model (Henderson and Venkatraman, 1993).	The paper addresses the question critical to practitioners: is your organization moving fast enough and in an appropriate way to keep pace with the changes in business transformation? The authors describe strategic alignment as a constant process and emphasize the processes, not the technology, in order to gain strategic advantage. This paper explains how the strategic alignment can be applied to any organization. Three major steps are identified in using the SAM: (1) identify the initial domain or perspective; (2) finding the best method (IT) for the perspective; and (3) establishing a procedure for moving the methods into the unaddressed domains. The authors further broke the alignment model into four distinct domains: strategy execution, technology potential, competitive potential and service level. Using these steps, managers are thought to continuously move through the four domains of the SAM, continually adjusting domains/perspectives based on changes in other areas. While this process is carried out, the authors suggest three things to keep in mind. Firstly, the first cycle through the SAM is the most difficult. Secondly, managers should continuously cycle – clockwise or counterclockwise – based on the initial flows. This will ensure that alignment is kept in balance. Thirdly, when diminished returns relative to the effort it takes to rotate through SAM are found, revise the number of times the organization cycles through the SAM.
Ma, Louis C.K. and Burn, J.M. (1998). *Managing the Dynamics of Information Systems Strategic Alignment. 1998 IRMA International Conference*, Boston, USA. 240.	Case study $N=74$ organizations	This paper utilizes STROBE – the Strategic Orientation of Business Enterprise model (Venkatraman 1985, 1989) and STROIS – the Strategic Orientation of IS model (Chan and Huff 1993; Chan, 1992)	This paper develops a configurational model on IS strategic alignment that evaluates not only the internal consistencies between IS strategy and business strategy, but also contingency approaches for different types of IS strategic alignment. The four alignment types are: (1) Business-strategy-led, (2) Conservative, (3) Organization-led, and (4) Technology-led. Quantitative assessments of survey data indicate that there is a relationship between alignment types and IS factors, as well as significant differences in IS strategic planning characteristics among the four alignment types. Qualitative analyses from the survey and multiple case study evaluations have identified good

Citation (Author, year)	Research method	Theory/concept	Findings
			practices and pitfalls to avoid in IS strategic planning, as well as the change processes in the migration from one alignment type to another.
MacDonald, H. (1991). The Strategic Alignment Process, in Scott Morton, Michael S. (ed.) *The Corporation of the 1990s: Information Technology and Organizational Transformation*. London: Oxford Press.	Conceptual	This author reviews the strategic alignment model (Venkatraman, 1991; Henderson and Venkatraman, 1989) and the elements of strategy.	This chapter first elaborates on the strategic alignment model by illustrating the strategic alignment process (SAP). This process acknowledges the fact that business strategy, and hence alignment, take place in the broader environment and that the broader environment can greatly impact alignment. The author then develops a matrix of business strategy and IT use based on exploitability (high, low) and interrelatedness (high, low). This matrix leads to four generic strategies: chaotic proliferation, preemptive penetration, barroom brawl, and clash of the titans. Each generic strategy has its own attributes, behaviors, and threats. Movement between the strategic quadrants tends to be induced by changes in the industry, technology, or in demands by customers. The author discusses business network redesign and business scope redefinition in detail. The chapter concludes with the author emphasizing the importance of alignment and the appropriate planning and processes in order to achieve alignment.
Madapusi, A. and D'Souza, D. (2005). Aligning ERP Systems with International Strategies, *Information Systems Management* **22**(1): 7–17.	Conceptual	The authors review the literature on ERP and strategic alignment. The authors also review the literature on international strategy types (multinational, global, and transnational). The authors also outlined the level of ERP configuration (enterprise level, system level, business process level, and customization level).	This paper looks at the alignment of enterprise resource planning systems (ERP) in large multinational enterprises. The authors first distinguished between large and small multinational enterprises, finding that there are many differences such as growth rates and organizational structures. The authors then developed a matrix for ERP alignment contrasting the international strategy and the level of ERP configuration. This matrix details the type and level of detail of the ERP as well as the type of roll-out of the system expect (phased or big bang). This matrix provides managers of large multinational enterprises with a framework to use when aligning ERP. The authors suggest five recommendations to managers and practitioners. Firstly, there is no one be all ERP solution. Success is the result of a carefully configured ERP system, a well-planned roll-out, and a judicious use of system capabilities. Secondly,

Citation (Author, year)	Research method	Theory/concept	Findings
			integrative thinking needs to be pervasive throughout the organization. Thirdly, always keep in mind that international strategy and ERP are intimately linked. Fourthly, appoint a senior executive to champion the ERP implementation and development. Finally, recruit employees with the necessary skills to integrate the ERP into the business strategy.
Maes, R. (1999). A Generic Framework for Information Management, Prime Vera Working Paper. Unpublished manuscript.	Conceptual	The authors extend the strategic alignment model of Henderson and Venkatraman (1993).	The author presents a generic framework used to coordinate different elements of IT. This model is an extension of the SAM model. The author first extends the SAM model vertically (the 'internal domain'), specifying it into a structural level and an operational level. The author then extends the SAM model horizontally adding a middle column labeled 'communication'. Communication is important in alignment as the use and sharing of internal and external information is necessary for alignment to occur, especially since most communication in the modern organization utilizes some form of technology. This extended model separates the business into three levels: strategy, structure, and operations. The strategic level recognizes that organizations need to ensure that their underlying technology and processes address higher level strategic issues. The structural level is important due to the emphasis on resource and core competence-based thinking. The operational level is important because it deals with skills and processes, and is essentially where one's strategy gets enacted. This model is embedded in the organizational environment.
Maes, R., Rijsenbrij, D., Truijens, O. and Goedvolk, H. (2000). Redefining Business–IT alignment through A unified framework. Universiteit van Amsterdam/Cap Gemini White Paper. Unpublished manuscript.	Conceptual	The authors extend the strategic alignment model of Henderson and Venkatraman (1993) and build off of the previous work done by Maes (1999).	This paper presents a generic framework for strategic IT alignment. The authors redefine alignment as 'the continuous process, involving management and design sub-processes, of consciously and coherently interrelating all components of the business–IT relationship', and states that it's a combined management-design problem. The model has three dimensions: main architecture areas (horizontal), design phases (vertical), and specific architectural viewpoints (depth). The mean architecture area includes business processes, information provision,

Citation (Author, year)	Research method	Theory/concept	Findings
			information systems, and technology infrastructure. The design phase includes the contextual, conceptual, logical, physical, and transformation phases. The authors regard their framework as a design tool, enabling management to position and interrelate different types of IS in order to achieve alignment.
McFarlan, W.E. (1984). Information Technology Changes the Way You Compete, *Harvard Business Review* **62**(3): 98–103.	Conceptual	Not applicable	This paper examines the ways that information technology changes the way organizations compete. The author demonstrates how IS can increase switching costs, build a barrier to entry, change the basis of competition, modify the balance of power between producers and suppliers, and generate new products. The challenge is that using IS strategy requires broad IS management, user dialogue, and a lot of imagination. Confounding these issues is the fact that IS investments are not easily measured. Yet the warning remains that it is much harder to play catch up once a competitor introduces new IS. The author offers five suggestions to practitioners. Firstly, the CEO must be able to articulate the IS developments in the organization and show where they will have an impact. Secondly, organizations must reverse the current industry norm of being open about there IS developments. Since IS is now strategic this information represents a competitive advantage and should not be shared. Thirdly, organizations should not use simplistic rules to calculate IS payoffs. Fourthly, organizations need to examine the second-effects of systems. The IS might have the desired impact in one area of the organization, but this may spill into other unexpected areas of the organization. Finally, organizations should not be too efficiency-oriented and should encourage creativity when it comes to IS.
McKenney, J., Mason, R. and Copeland, D. (eds.). (1996). *Waves of Change: Business Evolution Through Information Technology*, Cambridge, MA: Harvard Business School Press.	Book, various case studies	Not applicable	This book takes a historical look at the management approaches and strategies used to exploit the new developments in information technology. Of particular importance, the authors examine cases where the use of a particular technology in a given organization has completely altered the way that the entire industry operates. These industries now have what is termed a technological norm. The focus of the book is on

Citation (Author, year)	Research method	Theory/concept	Findings
			managerial actions, organizational change, and mutations in the use of technology. The following organizations are studied and their impact on their given industries is described: American Airlines, Bank of America, American Hospital Supply/Baxter Travenol, Frito-Lay, and USAA Insurance.
Miles, R.E. and Snow, C.C. (1978). *Organizational Strategy, Structure, and Process*, New York: McGraw-Hill Book Co.	Conceptual	The authors base their theory on three arguments: 1. Organizations act to create their environments; environmental enactment (Weick, 1969, 1977) 2. Management's strategic choices shape the organization's structure and processes. 3. Structure and process constrain strategy.	This book offers a theory of organizational strategic orientation and its relationships to structure and management processes. The authors outline the organizational process of adaptation, or the continuous and dynamic model of strategy. The authors address three adaptive problems: entrepreneurial, engineering, and administrative. The entrepreneurial problem typically occurs in new and/or changing organizations where the organizational domain must be defined. The engineering problem occurs when an organization must create a system to address the organization's entrepreneurial problem. Finally, the administrative problem involves addressing organizational challenges in a consistent way and in a way that will allow for continuance. These adaptive problems are intimately linked and organizations tend to progress by cycling through them. The authors also developed four organization types: defenders, prospectors, analyzers, and reactors. Defenders are organizations with highly narrow product domains and devote attention to improving efficiency in their current operations as opposed to seeking out new ventures. Prospectors continually seek out new market opportunities and create change and uncertainty in their industries; however, due to this uncertainty, they do not operate optimally. Analyzers operate in two domains – one that is stable and one that is scoped out for new opportunities. In the stable domain, operations are run efficiently whereas in the opportunistic domain, opportunities are acted on rapidly. Reactors are unable to respond to change effectively or to induce change in the industry. They make strategic adjustments when they are forced to do so and lack a consistent strategy–structure relationship.

Citation (Author, year)	Research method	Theory/concept	Findings
Miller, J. (1993). Measuring and Aligning Information Systems with the Organization, *Information & Management* **25**(4): 217–228.	Case study	The author also utilizes the premise that alignment of information systems with the organization is related to IS success (King 1978; Das *et al*. 1991; Henderson and Venkatraman, 1992). The author develops the Miller-Doyle instrument based on the previous research (Bailey and Pearson, 1983; Alloway and Quillard, 1981). The instrument, used to evaluate the overall IS functions, utilizes the perceptions of respondents regarding organizational importance and IS performance on a range of items.	The results are consistent with the proposition that user assessment of IS performance is influenced by the ability of IS staff to focus successfully on those aspects of IS that are regarded by IS as the most important. Then, IS staff is able to correctly interpret users and focus on user priorities. The study suggests that the popular approach of tapping user attitudes as a surrogate for IS effectiveness is not fully satisfactory – achievement of alignment between organizational priorities and IS performance is more fundamental. The author argues that in the era of all-pervasive application of information systems and technology, traditional methods of measuring the contribution and effectiveness of the information systems function are inadequate. The author points out that IS effectiveness criteria must recognize the dynamic nature of organizations, adapt to continuously changing organizational effectiveness criteria, and be influential in shaping management action, which in turn should be focused on improving alignment between IS and the business. This case study reveals the value of the presented instrument, which makes connection between IS capabilities and organizational purposes.
Mirani, R. and Lederer, A.L. (1998). An Instrument for Assessing the Organizational Benefits of IS Projects, *Decision Sciences* **29**(4): 803–838.	Instrument development, cross-sectional study of 178 IS projects	The IS effectiveness construct is based on several theories, in particular, the theory by Turner and Lucas (1985), and by Weill (1992). These theories were used to develop and operationalize a research framework.	The study delivers an instrument for measuring the organizational benefits of information systems projects. Benefits of this instrument are classified into three dimensions, namely (i) strategic, (ii) informational, and (iii) transactional. The following strategic benefits are identified: (i) competitive advantage, (ii) alignment and (iii) customer relations, where alignment benefits directly support organizational goals, help the organization create linkages with other organizations or enable the organization to reposition and be more responsive to environmental changes. The proposed instrument can be further utilized by researchers, who hypothesize and test the impact of various individual, IS and organizational variables on IS and organizational effectiveness. Based on the results, the authors suggest that any instrument applied to measure IS success should be tailored to the individual project under consideration. Only then can the assessment of the success of a specific IS project be meaningful.

Citation (Author, year)	Research method	Theory/concept	Findings
Ndede-Amadi, A.A. (2004). What Strategic Alignment, Process Redesign, Enteprise Resource Planning, and E-commerce have in Common: Enterprise-Wide Computing, *Business Process Management Journal* **10**(2): 184–199.	Literature Review and Conceptual	The authors focus on the relationships between enterprise-wide computing, enterprise resource planning, and strategic alignment.	This study serves as a retrospective and a rather broad view of former and current concepts of enterprise-wide computing systems, and their impact on integration of business processes. Successful organizations develop detailed analysis of interrelationships and data flows, and link their information needs into a fully integrated system development plan, which tie its system acquisitions to its business needs. Implementation of enterprise-wide computing system, which integrate the data created and used in different parts of the organization, helps organizations to achieve better control, coordination and governance. The results can be fostered by implementation of enterprise-wide computing systems in parallel with process redesign. The best rationale for acquiring IT is strategic alignment and the support of redesigned processes.
Niederman, F., Brancheau, J.C. and Wetherbe, J.C. (1991). Information Systems Issues for the 1990 s, *MIS Quarterly* **15**(4): 475–500.	Three-round Delphi survey. Round 1, $N=114$; Round 2, $N=126$; Round 3, $N=104$	Survey was composed of items deemed to be important taken from the IS literature and practitioner input.	This paper reports the findings of a survey of senior IS managers that asked them to assess which IS issues they felt would be most prevalent over the next 3–5 years. The top 10 issues identified by practitioners are: (1) developing an information architecture; (2) making effective use of the data resource; (3) improving IS strategic planning; (4) specifying, recruiting, and developing IS human resources; (5) facilitating organizational learning and use of IS technologies; (6) building a responsive IT infrastructure; (7) aligning the IS organization with that of the enterprise; (8) using information systems for competitive advantage; (9) improving the quality of software development; and (10) planning and implementing a telecommunications systems. IS organizational alignment was ranked #5 overall in the 1986 survey. IS organizational alignment was the seventh most important issue for IS practitioners and the fourth most important issue for IS observers.
Nolan, R.L. (1979). Managing the Crises in Data Processing, *Harvard Business Review* **57**(2): 115–126.	Conceptual	This article reviews the transitions that organizations are making due to the influence of information technology.	This article proposes a model for the six stages of data processing (DP) growth. The six stages of growth are initiation, contagion, control, integration, data administration, and maturity. These phases are marked by four growth processes: user awareness, DP planning and control, DP organization, and

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			applications portfolio. Managers can identify the various phases by looking at key benchmarks. At a more superficial, first-level analysis managers can look at the DP expenditure benchmarks and the technology benchmarks (i.e., what percentage of technology is focused on areas such as batch processing, data base processing, remote job entry, inquiring, time sharing, personal computing, etc.). A secondary level of analysis looks at the applications portfolio, DP organization, DP planning and control, and user awareness. The type of information system invested in also serves as a benchmark. Systems can either be used for strategic planning, management controls, and operational systems. In order to manage growth successfully, the author recommends five guidelines. First, recognize the fundamental transition from computer management to data resource management. Second, recognize the importance of enabling technologies. Third, identify that company's DP phase and track progress. Fourth, develop a multilevel strategy and plan. Finally, compose a steering committee to track progress and make the committee work.
Orlikowski, W. (1996). Improvising Organisational Transformation Over Time: A situated change perspective, *Information Systems Research* **7**(1): 63–92.	Case study, $N=1$	The author reviews the dominant theories of technology-induced organizational change: planned change, technological imperative, and punctuated equilibrium.	This article introduces the notion of situated change. Examining a case of a customer service department where a new electronic data collection method was introduced, the author outlines the changes that came about in the organizational members' behaviors. The new technology did not change behavior in and of itself. Instead the new technology created embedded constraints and enablers. This resulted in a series of metamorphic changes in the organization. The authors concluded that her results were contrary than what would be predicted by the punctuated equilibrium theory of change – small changes over time can result in organizational metamorphoses. The key features of this perspective are that the technology in use needs to be flexible and constantly updated to meet changing needs (as opposed to be too rigid). Also, customization is always needed. This perspective views technology and organizational change as ongoing.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Palmer, J.W. and Markus, M.L. (2000). The Performance Impacts of Quick Response and Strategic Alignment in Specialty Retailing, *Information Systems Research* **11**(3): 241–259.	Case study (based on the cross-sectional survey of specialty retailers), $N=80$	Quick Response technology consists of four levels of successively more sophisticated technologies and applications (Retail Information Systems News, 1993, 1995), positioned as 'array of technology options for supporting the company's mission' (Retail Information Systems News, 1996). In this paper, the authors define strategic alignment as the correlation between the business strategy of a firm and the firm's IT strategy. Since the authors are concerned with the effect of the two co-aligned variables on a third (i.e. performance), they use the definition of alignment as moderation (Venkatraman, 1989). The performance construct include key operating ratios used in the specialty retailing industry, which include profitability, sales growth, sales per employee, sales per square foot and stock returns. These performance measures are used by experienced industry analyst e.g., Standard and Poor's (1993). Profitability, as a standard measure, is calculated by dividing net income by sales.	Specialty retailers are companies occupying a unique position in the retail pyramid, and which adopt a focused approach to retailing (Standard and Poors, 1993). The focus may lie in product offerings, product delivery or customer service. Therefore, quick response (QR) technology may represent a competitive advantage. The results reveal significant positive associations between QR use and profitability, comparable store sales, sales per square foot, and stock turns, but not sales per employee. On that last measure, non-adopters of QR outperform adopters. Thus, the findings strongly support the hypothesis that in specialty retailers, adopting QR technology is associated with increased firm performance compared to non-adoption. The findings indicate two types of the match in the alignment matrix, namely (i) firms with both an internal business strategy focus and a transaction efficiency focus in their IT, and (ii) firms with both a customer strategy focus and a customer detail IT focus. These findings support the hypothesis that specialty retailers adopt an IT strategy consistent with their corporate strategy. The results show that QR adoption is the best predictor of most performance measures. Although strategic alignment is a common practice among surveyed companies, alignment is not significantly associated with organizational performance. The study proves existence of an association between adoption of the QR program and several measures of performance, and an apparent match between IT and business strategy. Adoption of the QR program at a minimal level is associated with higher performance, although there is no performance impact due to higher levels of QR use. The authors imply that the association of QR adoption with higher performance is that higher performing firms are better able to afford QR technologies. The findings suggest the need to redefine the QR program and the QR hierarchy. The study represents more convincing evidence against strategic alignment theory than do prior studies. For instance, the results differ from research by Papp and Luftman (1995) and Luftman (1997), who found few firms in alignment, as

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			opposed to this study, which identifies many firms. The result support prior work showing that a company's investments in transaction-oriented IT improve organizational performance (Weill, 1992).
Papke-Shields, K.E. and Malhotra, M.K. (2001). Assessing the Impact of the Manufacturing Executive's Role on Business Performance Through Strategic Alignment, *Journal of Operations Management* **19**(1): 5–22.	Survey, $N=202$ with follow up interviews	This model is based on the literature around procedural justice and strategic information systems management.	This research looks at the impact that the manufacturing department can have on business strategy. This impact is hypothesized to take place via the strategic alignment process. The involvement and influence of manufacturing executives was found to be important in this study, although they are distinct concepts. Involvement and influence in the business strategy process led to greater alignment and hence performance. However, the degree of involvement and influence is not under the control of the individual, but rather the organization (climate, structure, etc.). Involvement and influence can exist together or can lead to one another in a given scenario (i.e., I became involved because I had influence). However, influence plays a more substantive role in this relationship. The authors suggest for future research to look at the impact of environmental uncertainty in a given functional area and its impact on involvement and influence.
Papp, R. (1999). Business-IT Alignment: Productivity Paradox Payoff? *Industrial Management+Data Systems* **99**(8): 367–373.	Longitudinal (5 years) case study $N=500$ firms	The authors apply the strategic alignment model (Henderson and Venkatraman, 1989, 1992, 1993).	This paper addresses the need for the development of some sort of financial measure of alignment. The author defines the following 7 financial measurements of alignment impact on financial performance: anticipated performance, liquidity, income, growth, net profitability, earning and debt-to-equity. Since alignment remains one of the leading areas of focus among business executives, the author suggests that management should take the following steps to achieve alignment as a means of improving performance and profitability: • Assessing the firm's perspective using the alignment model. • Learning to recognize and leverage IT within your firm to maximize efficiency. • Incorporating financial measurements suitable for your particular industry when assessing alignment.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			• Giving everyone in the firm a clear and useful role to facilitate synergy between IT and the business. • Never stopping the assessment of alignment within the firm. It is a continuous, dynamic process requiring constant monitoring.
Peak, D., Guynes, C.S. and Kroon, V. (2005). Information Technology Alignment Planning – A case study, *Information & Management* **42**(5): 635–649.	Case study: one site; Omaha Public Power District; 58 managers from five major business units	This paper uses the Information Technology (IT) Alignment Planning Process. Measures of Information Quality (IQ) were developed. There are four dimensions of IQ: critical success factors (CSF); business processes, information needs, and IT products and systems.	This paper develops a step-by-step process to assess Strategic IT alignment in an organization. The model looks at how critical success factors, IT products/systems, information needs and business processes interact with each other. The IT strategic plan is built by an organization's information technology alignment plan, which is a combination of each BU's IT alignment roadmap. A BU's IT alignment roadmap is the result of a BU's current IT solution and activity, combined with identified information gaps. A key contribution from this study was the use of color-coded alignment assessments that was intuitively pleasing to management (easier to identify problems with information when looking at colours instead of numbers).
Pearlman, E. (2004, October 15, 2004). Welcome to the Crossroads, *CIO Insight* **1**(45)	Interview $N=2$	This article reports on an interview with Marianne Broadbent and Ellen Kitzis who expanded on some of the concepts presented in their book, *The New CIO Leader: Setting the Agenda and Delivering Results* (Harvard Business School Press, Dec. 2004), and put them in the context of corporate culture.	According to Broadbent and Kitzis, these are the 10 priorities the New CIO Leader must have (with more details on the alignment related points): 1. Lead, don't just manage. 2. Understand the fundamentals of your environment. 3. Create a vision for how IT will build your organization's success. You must have a vision for achieving your colleagues' business goals using technology. 4. Shape and inform expectations for an IT-enabled enterprise. You need to work with your colleagues to identify the key business needs, strategies and drivers, and then articulate the IT guidelines necessary to address those needs. 5. Create clear and appropriate IT governance. Effective governance enables you to weave together business and IT strategies and to consistently build credibility and trust. 6. Weave business and IT strategy together. IT strategy means developing and actively managing

Citation (Author, year)	Research method	Theory/concept	Findings
			your IT portfolio to deliver success as measured by your colleagues. 7. Build a new IS organization – one that is leaner and more focused. 8. Build and nurture a high-performing team in your IS organization. 9. Manage the new enterprise and IT risks. 10. Communicate IS performance in business relevant language. You must know and communicate how IT is contributing to shareholder value and the IT value indicators that are directly linked to business value measures.
Peppard, J. and Breu, K. (2003). Beyond Alignment: A coevolutionary view of the information systems strategy process. Twenty-Fourth International Conference on Information Systems, Seattle, USA, 743.	Conceptual	This paper utilizes the coevolutionary perspective as a new lens for viewing alignment. Central to this are the notions of adaptation and change.	This paper addresses the question of how to achieve and sustain alignment over time and proposes the coevolution theory of strategic IT alignment. Coevolutionary theory does not view the organization or process in isolation; it views the organizational as part of a larger socio-cultural and historical system. Fundamental to this perspective include uncertainty, complexity, munificence, graininess, fitness, and niches (McKelvey, 1997) as well as absorptive capacity, fit with the environment, and value creation mode. This perspective also recognizes that actions of one firm will have an impact on both other firms and the environment. Thus, the coevolutionary process allows researchers to view alignment as a fuzzy, indeterminate process. The authors create a coevolutionary model of strategic IT alignment. In order for business and IT strategy to align, the following factors must be at play: structural alignment, fitness function, path dependency, absorptive capacity, landscape (rugged *vs.* smooth), value creation (exploration *vs.* exploitation), technology (emerging *vs.* disruptive), competitive dynamics, and strategic alignment. A change in one of these variables has multilevel effects on the other variables. As this paper represented research-in-progress, the authors suggest that future research over a longer time horizon is need to assess the benefits of the coevolutionary theory of IT alignment.

Citation (Author, year)	Research method	Theory/concept	Findings
Pollalis, Y.A. (2003). Patterns of Co-Alignment in Information-Intensive Organizations: Business performance through integration strategies, *International Journal of Information Management* **23**(6): 469–492.	Survey $N=183$ banks	The Gestalt approach suggested by organizational and strategic management theorists (Mintzberg, 1979; Miller and Friesen, 1978; Miller, 1982, 1986) was a base of holistic methodology used in this research.	The findings in this paper indicate that banks with consistent levels of strategic alignment process and output will perform much better than other similar banks when they have a high degree of TI (TI-output) and IT-based functional coordination (FI-output) in place. The findings here also imply that (a) outsourcing could affect performance positively as long as the implementation of IT integration is consistent with the needs for IT-based functional coordination, (b) banks with inconsistent strategic alignment levels and low levels of TI and FI output integration will perform poorly compared to the rest of the banks, (c) small banks with high levels of TI and SI are, in general, good performers, and (d) large banks with low or medium levels of TI and SI are, in general, poor performers.
Powell, T.C. (1992). Organizational Alignment as Competitive Advantage, *Strategic Management Journal* **13**(2): 119–134.	Survey, $N=113$	The central interest in this research is a proposition that organizational performance is a function of the fit between an organization's structural differentiation and integration (Lawrence and Lorsch, 1967).	This research attempts to integrate the strategic management and organizational contingency perspectives. The results support the notion that some organizational alignments generate extraordinary profits, which are not connected to the profits produced by traditional industry and strategy variables, and that organizational factors can act as sources of competitive advantage. The authors concluded that structural fit explains a significant increment of profitability variance, which acts as a source of competitive advantage independent of industry and strategy content. The results reveal that two of the generic strategy variables, namely production costs and market breadth, correlate significantly with firm's profitability, while firm age correlate negatively with profitability. The article suggests that by focusing on industry and competitive strategy variables, industrial organization and strategy research has diminished the role of organizational factors in producing sustainable competitive advantage.
Pyburn, P.J. (1983). Linking the MIS Plan with Corporate Strategy: An Exploratory Study, *MIS Quarterly* **7**(2): 1–14.	Case study $N=8$ organizations	The strategic alignment planning process is studied. Planning is a developmental process rather than an observable outcome or document. Strategic IS planning is concerned with the relationship between the overall organization's missions,	This article looks at the organizational planning processes that are important to organizations. The following factors were observed: • IS is much more likely to be perceived as supporting the critical needs of the business when there are few levels between the senior management and the IS manager.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
		objectives and strategies, and the organization's MIS. IS strategy emanates directly from corporate strategy with additional focus on the linkage between the two. Linkage between IS strategy and corporate strategy, which is the province of MIS planning. Effective strategic planning is a process of building some consensus regarding the role of IS *vis-à-vis* the rest of the firm and the resources committed to achieving that role.	• The influence of an IS manager's status on the success of IS planning is independent of the style of that planning. On the other hand, the more informal the style, the more important this status becomes. • IS groups which are located many miles from headquarters will require more formal IS planning procedures than those which are close and/or in-house. • In the environments with informal personal relationships between the IS manager and senior management, the formal IS planning does not appear to be a critical determinant of perceived IS effectiveness. • As organizations become more formal in their business practices, the role of formal strategic IS planning disciplines become more significant as a countervailing force. • As organizations grow in size and complexity, more formal planning processes help to ensure an integrated vision for IS. • The success of the personal-informal style of strategic planning seems to depend in part on both an informal general management style and high IS manager status. • The personal-informal style seems most appropriate when the business environment is rapidly shifting and dynamic. • With the personal-formal and written-formal planning styles, success depends in part on a more formal general management style. • The utilization of a number of forms, written documents and formal planning structure of the make it much easier to integrate across complex system activities. • The implementation of any IS planning process must consider the several possible avenues for developing the mutual understanding so essential in planning.
Raymond, L., Pare, G. and Bergeron, F. (1995). Matching Information Technology and Organizational Structure: An empirical study with implications for performance,	Survey, $N = 108$	The author utilizes the notion of sophistication of the organization's use and management of IT (Cragg and King, 1992), the contingency theory in IS research (Iivari, 1992)	The study results reveal that IT sophistication is positively related to structural sophistication, IT usage is positively related to organizational performance, and the relationship between IT management and structural sophistication is stronger among the better-

Citation (Author, year)	Research method	Theory/concept	Findings
European Journal of Information Systems **4**(1): 3–16.		and the framework for the organizational impact of IT (Bakos, 1987). The research model hypothesizes relationships between the sophistication of IT, the sophistication of the organizational structure and organizational performance.	performing firms as compared to worst-performing firms. The role of the environment is fundamental. Better performance results when operating in less uncertain environment with neither size nor the structural factors contributing to that performance outcome. However, the IT usage, which is independent from structure, proved to have significant effects on performance. Sophisticated firms use more diverse technologies in support of key production functions, which include management support applications, as opposed to simple transactional functions.
Reich, B.H. and Kaarst-Brown, M.L., (2003). Creating Social and Intellectual Capital Through IT Career Transitions, *Journal of Strategic Information Systems* **12**(2): 91–109.	Case study of one organization (Clarica Life Insurance Company). Data were collected over a period of 3 years with 70 former IT employees who transitioned within the company.	The authors utilized Nahapiet and Ghoshal's theory of co-creation of social and intellectual capital. Social capital creates the environment in which intellectual capital can be created. This leads to organizational advantage that further increases social capital. This is a virtuous circle. This lens has ties to Burt's (1992) structural holes. If IT people are transferred out of the IT department, they become a network spanner and increase their social capital.	This paper looks at the benefits of transferring IT employees to other non-IT departments in terms of intellectual and social capital. The findings include that transferring IT people led to an increase in intellectual capital in the firm (in terms of IT knowledge). This led to organizational advantage in terms of retention of IT people and IT innovation, which in turn led to an increase in social capital (stronger IT-line relationships) and recruiting advantages. The transition of IT personnel into line positions was facilitated by the company's loyalty to IT and to the business literacy of the IT professionals.
Reich, B.H. and Benbasat, I. (2000). Factors that Influence the Social Dimension of Alignment Between Business and Information Technology Objectives, *MIS Quarterly* **24**(1): 81–113.	Case study, $N=10$ business units at life insurance companies. Unstructured interviews, $N=57$, from 45 informants	The social dimension of alignment is derived from the political behavior model, resource dependency theory, and the social construction of reality.	This research investigates the social dimension of alignment. It looks to identify factors that either enable or inhibit alignment. The proposed model states that a shared domain of knowledge between business and IT executives and successful IT history lead to increased communication between business and IS executives and connections between business planning and IT planning. Altogether this leads to greater alignment. This model was significantly supported for short-term strategic IT alignment. When looking at short-term alignment, a fifth factor was found to influence alignment directly – short-term business direction. The most important factoring influencing short-term alignment was the frequency of unstructured or structured communication. However, when looking at long-term alignment an interesting model presented

Citation (Author, year)	Research method	Theory/concept	Findings
			itself. Shared domain of knowledge between business and IT executives and long-term business direction led to alignment. The findings indicate that both researchers and practitioners should devote great effort towards understanding shared domain of knowledge.
Reich, B.H. and Benbasat, I. (1996). Measuring the Linkage Between Business and Information Technology Objectives, *MIS Quarterly* **20**(1): 55–81.	Case study $N = 10$ business units at three major insurance companies	Authors suggest linkage (e.g. alignment) can have social and intellectual dimensions. They consider social alignment to be a state of mutual understanding and commitment between IT and business strategies and plans. Using previous research, the authors develop several potential measures of the social dimension of alignment and use 10 case studies to evaluate them.	The study validated two out of four proposed measures of linkage, namely understanding of current objectives and congruence in IT vision. The third measure, self-report, can be utilized if administered appropriately. The study also verified short and long-term aspects of linkage. The measures can be used to perform a linkage audit, which can assess the level and types of linkage. The study revealed that presence of factors, which might lead to linkage, can mislead executives into believing that linkage exists. Contrary to normative literature, both short- and long-term linkage were not required to be present at all times.
Reich, B.H. and Huff, S. (1991). Customer-Oriented Strategic Systems, *Journal of Strategic Information Systems* **1**(1): pp. 29–37	Case study of 11 first-mover strategic information systems in Canadian companies	The study investigates Customer Oriented Strategic Systems (COSS), which are those Strategic Information Systems (Wiseman, 1985), which link a company with its customers. The research extends the study of strategic systems by Runge (1985). The article proposes a 3-stage model of strategic systems impact: Implementation, early adoption, final adoption. The factors influential to each stage are shown to differ from one another.	The results support previous research in the implementation of information systems and strategic systems literature, namely (i) factors related to successful implementation of IS and the competitive environment were associated with systems that were developed and introduced to the market first, (ii) factors related to the adoption of innovations and IS, and to successful product marketing were associated with high adoption. In addition, the study reveals that (i) poor support for the sales force and poor quality pilot test inhibit early adoption of the system, (ii) lost of direct control over COSS inhibits long-term penetration, and (iii) if systems does not achieve high long-term adoption or are present less than 3 years in the market, competitive advantage is not achieved.
Reponen, T. (1998). The Role of Learning in Information Systems Planning and Implementation, in R.D. Galliers and Baets, Walter R.J. (eds.) *Information Technology and Organizational Transformation: Innovation for the 21st Century*	Conceptual/ Literature Review	This chapter is founded on the Evolution Model for Information Systems Strategy (EMIS) (Reponen, 1989).	This chapter aims to help management make IS plans that are well-suited to the organization's situation and objectives. The authors use the term strategic information systems planning (SISP). In order to do so, those planning and implementing the IS strategy require both business skills and IS skills. However, a gap often exists where business managers lack IS skills and knowledge, and IS managers lack business

Citation (Author, year)	Research method	Theory/concept	Findings
Organization, 2nd edn., Toronto: John Wiley & Sons, pp. 133–149.			knowledge and skills. To address this gap, the authors emphasize the notion of learning. The importance of learning has been emphasized in the interactive planning process literature. Of particular importance is awareness, the desire to participate on the part of business executives, interaction between business and IS executives, learning and the desire to learn, attention to human type and personality in order to optimize the team process, having a project champion, formulating a decision making strategy, and having the appropriate technology to support the IS strategy. With the appropriate IS planning system in place organizations can use IT to their strategic advantage.
Rockart, J.F., Earl, M.J. and Ross, J.W. (1996). Eight Imperatives for the New IT Organization, *Sloan Management Review* **38**(1): 43–55.	Literature Review/ Conceptual/Prescriptive	The authors reviewed the current state of the business world including issues of reengineering and technological change.	This article outlines eight key IT factors that a business needs to master in order to keep up with the fast-changing business environment. The changing environment has led to four major process changes: reengineering operational processes, reengineering supporting processes, rethinking managerial information flow, and redesigning network processes. Technological change in the form of distributed computing environments, new development methods, exploding technology, new industries, and the availability of outside suppliers, has added to the pressure to use IT more effectively. The eight imperatives to the new IT organization are: (1) achieve two-way strategic alignment; (2) develop effective relationships with line management; (3) deliver and implement new systems; (4) build and manage infrastructure; (5) reskill the IT organization; (6) manage vendor partnerships; (7) build high performance; and (8) redesign and manage the federal IT organization. The authors elaborate on two-way strategic alignment as meaning that just as the business strategy should influence the IT, the IT should influence the business strategy. This influence should go both ways. In order to achieve these new imperatives, a few new core activities must be performed. The new IT needs to be more of an advisory, management, and business-centered function. Its two core activities are working with line managers and ensuring that the organizational strategy is supported and driven by IT.

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Roepke, R., Agarwal, R. and Ferratt, T.W. (2000). Aligning the IT Human Resource with Business Vision: The leadership initiative at 3 M, *MIS Quarterly* **24**(2): 327–353.	Case study $N=1$ company (3 M)	The authors look at human resource issues in IT. Specifically, they examine the impact that leadership has on alignment. This research also examines the unique culture produced at 3 M.	The results indicate that leadership capability in all IT professionals is a potential source of sustainable competitive advantage, and thus, investment in a leadership capability can help align IT with the business vision and provide value to IT customers. However, the authors point out that 3M's approach to alignment seems to be unique. Alignment of individual needs and values with organizational goals is an essential component of recruitment and retention success, which extends the typical notion of strategic alignment for the IT function that excludes human resource challenges. The leadership has two alignment responsibilities, namely (i) ensuring that all work groups are focused on the right things to do, and (ii) developing a clear understanding of the passion and competencies of every individual in their work group. The study identifies leadership capabilities in all IT professionals as a source of sustainable competitive advantage. It suggests an approach for developing a leadership capability, which includes: • a transition model, which describes how IT managers and IT professionals should interrelate; • a personal leadership model, which includes behaviors of initiative, emotional self-management, cooperation, customer service orientation, self-confidence, achievement orientation, flexibility, and personal understanding; • a model and curriculum helping IT professionals achieve alignment with their customers and within a work group; • a shared leadership architecture to incorporate above elements in a broader model of leadership development
Rondinelli, D., Rosen, B. and Drori, I. (2001). The Struggle for Strategic Alignment in Multinational Corporations: Managing readjustment during global	Case study $N=1$ MNC, interviews and archival data	Strategic alignment and planning theories suggests that MNCs should continuously readjust and realign four sets of strategic components — business strategy, market penetration decisions, management processes,	A new challenge is maintaining alignment while expanding internationally at a fast pace. This article explores the concepts and components of strategic alignment in MNCs by analyzing PlantNutrients International (PNI), a US-based privately owned holding company that grew rapidly during the 80 s

Citation (Author, year)	Research method	Theory/concept	Findings
expansion, *European Management Journal* **19**(4):404–416.		and organizational structures — as they expand internationally.	and 90 s. The findings were that while the academic literature on strategic alignment suggests a rational and deliberate process, this case analysis indicates that achieving alignment is chaotic, complex, and frequently imperfect. The study derived a set of insights or lessons learned about alignment from their analyses that may prove useful to MNC managers and may guide future researchers interested in corporate alignment: alignment is not always rational and systematic; alignment is a proactive process; alignment is a painful process; alignment is a continuous process; alignment requires buy-in; and alignment begins with strong leadership.
Ross, J., Beath, C. and Vitale, M. (1996). Develop Long-Term Competitiveness Through IT Assets, *Sloan Management Review* **38**(1): 31–45.	Conceptual, reviews case studies	This article utilizes the resource-based view of the firm.	This article looks at how IT can be developed to a competitive capability. The authors identify that IT is a competitive capability when firms can control IT-related costs, deliver systems when needed, and shape the business through IT implementations. The authors propose a framework to develop this capability. This framework contains a relationship, human, and technology component. These factors represent a highly competent IT staff, a reusable technology base and a strong working relationship between the IT and business executives. These factors help the organization perform better as they build and leverage IT processes such as strategically aligned IT planning, fast delivery, and cost-effective operations and support. The authors also develop a matrix where companies are placed based on their competitive environment (immediate threat, no immediate crisis) and their IT asset (strong, weak). This leads to four categories of IT capabilities: sinking, drifting, luffing, and cruising. The authors warn organizations to never cruise. In order for IT to remain a competitive advantage it must be used and refined.
Sabherwal, R. and Kirs, P. (1994). The Alignment Between Organizational Critical Success Factors and Information Technology Capability	Case study, $N=244$ large academic institutions in the US	The study examines performance implications of alignment between Critical Success Factors (CSFs) and IT capability. Alignment is measured using profile deviation, and is	This article investigates the relationship between critical success factors and one organizational competence: IT capabilities. The study suggests that alignment has performance implications, namely it was positively associated with both perceived IT success

Citation (Author, year)	Research method	Theory/concept	Findings
in Academic Institutions, *Decision Sciences* **25**(2): 301–330.		considered to be the opposite of misalignment; the misalignment measure, in turn, is calculated as the weighted Euclidean distance of the IT capability variables from the ideal profile.	and organizational performance. The authors suggest, however, that organizations with higher levels of perceived IT success might recognize the importance of IT capability and thus, strive harder to align it with CSFs. The results support the following hypotheses (i) that organizations operating in uncertain environments may be able to derive greater benefits from IT, (ii) organizations which have more sophisticated/advanced IT management obtain greater benefits from IT, (iii) organizations with more sophisticated or advanced IT management have closer alignment between CSFs and IT capability.
Sabherwal, R., Hirschheim, R. and Goles, T. (2001). The Dynamics of Alignment: Insights from a punctuated equilibrium model, *Organization Science* **12**(2): 179–197.	Case study $N=3$ organizations	This paper uses the punctuated equilibrium model to study strategic IS management, which argues that periods of gradual evolution are 'punctuated' by sudden revolutionary periods of rapid change (Elderidge and Gould, 1972; Van de Ven and Poole, 1995).	This paper advances our understanding of the dynamics of alignment. The paper suggests that claims about performance effects of alignment should be couched in explicitly longitudinal terms because the same alignment pattern may not be effective over extended periods. Based on the application of the punctuated equilibrium model to the three cases, the paper suggests that changes in alignment are, for the most part, small and evolutionary. These changes prevent catastrophes by controlling misalignments, but they inhibit moving to an altogether different pattern of alignment.
Sabherwal, R. and Chan, Y.E. (2001). Alignment Between Business and IS Strategies: A Study of Prospectors, Analyzers, and Defenders, *Information Systems Research* **12**(1): 11–33.	Case study, based on two surveys $N=164$ and 62 companies with a 4-year time gap between them.	The paper combines two approaches to measuring business strategy- the Miles & Snow typology (1978) and STROBE (Venkatraman, 1989). The paper examines alignment between a company's actual IS strategy and the theoretically predicted IS strategy.	This paper examines alignment using Miles and Snow's (1978) classification of business strategy. It examines alignment using a broad, holistic perspective. The study identifies profiles of IS strategy which are most suitable for each business strategy defined by Miles and Snow. Alignment is seen to be significantly correlated with perceived business performance, with the nature of the relationship being complex and depending on the business strategy. The study demonstrates a practical approach to integrating Miles and Snow's business strategies with Venkatraman's STROBE approach (Venkatraman, 1989).
Sauer, C. and Yetton, P.W. (1997). The Right Stuff – An introduction to new thinking about management, in	Literature Review	This chapter reviews the relevant literature from strategic alignment and IT. This is an introductory book chapter.	Two main approaches have become popular since mid 1980 s, namely (i) IT strategic planning (Ward, Griffiths and Whitmore, 1990), and (ii) organizing to

Citation (Author, year)	Research method	Theory/concept	Findings
C. Sauer and P.W. Yetton (eds.) Steps *to the Future: Fresh Thinking on the Management of IT-Based Organizational Transformation*, 1st edn., San Francisco: Jossey-Bass, pp. 1–21.			achieve strategic alignment (Henderson and Venkatraman, 1992). The basic principle of strategic alignment is that IT should be managed in a way that mirrors management of the business. Since 1980 s, when organizations realized the strategic role of IT, management has faced a challenge to move the organization from their current competitive situation to a position where IT takes on a new strategic role a new way of doing business. However, this is not a complete solution. Corporations may still have performance problems and a management problem with IT. Thus, organizational change needs to be undertaken in order for an organization to compete with the new information systems.
Sauer, C. and Burn, J.M. (1997). The Pathology of Strategic Management, in C. Sauer and P.W. Yetton (eds.) *Steps to the Future,* 1st edn., San Francisco, CA: Jossey-Bass.	Case study, $N=4$	This chapter examines the fundamental theoretical principle of strategic alignment (and misalignment) to demonstrate that the potential for pathological outcomes is firmly embedded in the principle of alignment itself.	The main purpose of this chapter is to sound a warning that alignment by its very nature gives rise to pathologies that may not always be avoidable; therefore, careful management is required to avoid undesired business and IT costs. Three types of pathological outcomes from strategic alignment are identified: (i) misalignment, which occurs when a company tries to align IT with the business but the business is not in a state of tight fit; (ii) IT stagnation, which occurs as part of a common, almost natural, cycle of innovation; and (iii) globalization, which presents special difficulties for alignment. The authors conclude that the pathologies observed are inherent in alignment rather than the result of some contingent shortcoming. The root of these pathologies is the deep assumption that the business and IT should be kept separate.
Scanlon, J. (2003, January 8, 2003). A Capital One Exec Addresses Alignment, *CIO Insight* **1**(22)	Interview, $N=1$ executive	This article reports on an interview with John Scanlon, recipient of CIO Insight's *Partners in Alignment Award* for his company, Capital One Financial Corp.	Scanlon suggests that we need to find ways to build bridges so that both IT and business personnel are using the same language and talking about the same topics. This will help get alignment in thought, action, and practice. Alignment has to be more of a cultural phenomenon, where every single associate in the IT organization cares about alignment and tries to work the practices of alignment into how they do their job, including the business partners. Organizations should not wait for business partners to lead this effort. The strategic

Citation (Author, year)	Research method	Theory/concept	Findings
			planning process that business partners use does not give IT partners the information they need to do technology planning. Finally, Scanlon suggests that practitioners do not try to prove alignment since it's kind of an unprovable thing. Alignment is a never-ending quest.
Scarbrough, H. (1998). Linking Strategy and IT-Based Innovation: The Importance of the "Management of Expertise", in R.D. Galliers and Walter R.J. Baets (eds.) *Information Technology and Organizational Transformation: Innovation for the 21st Century Organization,* 2nd edn., Toronto: John Wiley & Sons, pp. 19–36.	Case Study, $N=3$ comparisons	The author utilizes social networks theory and reviews the classical and processual approaches to strategy.	This chapter examines the role of expertise of business executives in the management of IS. The author questions the notion that IT is invested in and implemented due to rational thoughts. Instead, it is proposed that the management of expertise perspective is concerned with the counter-factual. It explains why two organizations with similar situations manage IT in completely different ways. The author looks at three paired case studies where similar situations led to completely different IT outcomes. The author suggests that there is no rational explanation, but rather top management's expertise of IT and its potential to innovate came into play. If management was aware of IT and the potential innovations, then IT investments were readily accepted. If they did not have this expertise, political and power struggles emerged.
Shank, J.K., Niblock, E.G. and Sandalls, Jr. W.T. (1973). Balance Creativity and Practicality in Formal Planning, *Harvard Business Review* **51**(1): 87–95.	Case Study, $N=6$	This study reviews the corporate planning process.	This article looks at the formal planning activities that take place in an organization. The authors cite the tension between balancing practicality and creativity, both of which are important in creating and implementing a successful strategy. The author state that practitioners should aim for 'realistic creativity'. They also state that it is the corporate objectives that dictate the organizational planning system (not the other way around).
Sledgianowski, D. and Luftman, J. (2005). IT-Business Strategic Alignment Maturity: A case study, *Journal of Cases on Information Technology* **7**(2): 102–120.	Case study: one organization (specialty chemical manufacturer)	The Strategic Alignment Model (Luftman, 2000) was as the framework. A case study was used case study to develop 'key insights' and 'SAM best practices' under the headings: communications; competency/value management; governance; partnership; scope and architecture; and skills.	Major Insights: 1. Interorganizational communication is fostered by a culture that promotes regularly occurring communication as a fundamental task of every manager and employee (communications) 2. Include business-related metrics, such as user satisfaction and IT's responsiveness to the business, with technical service level agreements (SLA), such as computer response time and minimum downtime, to help for more of a partnership between IT and

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			business. Additionally, measurements like contribution to profits, quality, and productivity improvements should be applied whenever possible (competency/value measurement) 3. Use global champions to both locally promote IT initiatives and act as global liaisons. (partnership) 4. A single one-company solution with common processes, data and systems provides a common language to facilitate common understanding across countries, functions, and segments resulting in significant cost savings for both Customer Order Desks and IT. (scope and architecture) 5. Provide multiple methods for development of IT and business managers, including methods of on-the-job training, job rotation, job enrichment, and international assignment. (skills) SAM Best Practices: 1. Communication between IT and business should be pervasive throughout the organization, informal, regularly occurring, and use rich methods such as email, videoconferencing, and face to face (communications) 2. Have periodic formal assessments and reviews of SLAs with both IT and business representation to mandate continuous improvement of SLAs and the attainment of business objectives. (competency/value measurement) 3. Have frequent and formal assessments and reviews of IT investments with a formal process in place to make changes based on the results of the assessments. When possible, include business partners in the process. Share the knowledge gained across the organization. (competency/value measurement) 4. Existence of an IT steering committee increases the alignment of IT and business strategies. (governance) 5. IT should be a teammate with the business that co-adapts and improvises with their business partners in bringing value to the firm and meeting strategic objectives. (partnership) 6. Leverage IT assets on an enterprise-wide basis to extend the reach (the IT extrastructure) of the organization into supply chains of customers and suppliers (scope and architecture).

Citation (Author, year)	Research method	Theory/concept	Findings
Smaczny, T. (2001). Is An Alignment Between Business and Information Technology the Appropriate Paradigm to Manage IT in Today's Organisations? *Management Decision* **39**(10): 797–802.	Conceptual	The authors review the strategic alignment model (Henderson and Venkatraman, 1993) and the process views of alignment.	This article investigates the following research question: is the alignment between business and IT the appropriate paradigm to manage the information technology functions in organizations today? The author questions the implicit assumptions about alignment and states that due to the increased turbulence and change in the organizational environment, strategic IT alignment no longer holds. They describe the world as more 'react and respond' as opposed to careful planning. The authors propose a practical alternative to SAM. The author suggests that fusion – or when IT and business are one –be considered as an alternative to SAM. There are some key differences between alignment and fusion. In fusion there is only one strategic plan and one set of operational plans, as opposed to separate plans for both business and IS. The author calls for future research on the topic of fusion.
Smith, J.H. and Heinrichs, W. (1998). Organizational Alignment Through Information Technology: A web-based approach to change, ACM Conference, Seattle, USA.	Case study, $N=1$ (NASA)	The authors based their research on the web-based model	The authors presented a case study of a NASA process that was streamlined in order to be more efficient. The original process was not integrated and very time consuming. In order to process an order numerous people had to pass around the order over email, hard copies, and voice mail. The new integrated, wed-based process saved significant time and led to increased customer value. All form entries were made electronically. This case study serves as an example of a new process that increased IT alignment.
Soh, C., Kien, S.S. and Tay-Yap, J. (2000). Cultural Fits and Misfits: Is ERP a Universal Solution? *Association for Computing Machinery: Communications of the ACM* **43**(4): 47–51.	Case study, $N=7$ hospitals	The authors reviewed the research performed on ERPs.	A common problem with ERPs is misfits – or parts of the software that do not exactly fit with the organization. This article examines the notion of misfits from a cultural perspective. Namely, ERPs are designed with North American and European processes in mind and the authors investigate the implementation of ERPs in Asia. Misfits arise in the data, functionalities, and in the output. When a misfit occurs, the organization has four options: adapt the ERP, accept the shortfall in functionality, workaround the misfit, and customize.

Citation (Author, year)	Research method	Theory/concept	Findings
			The ERP implementation in Asia was generally positive. The authors conclude that ERP vendors need to spend more time incorporating embedded organizational processes into ERPs. Secondly, the process-focus of the ERP tends to ignore the data requirements. Finally, the three users of the ERP – key users, IS department personnel and the vendor – need to work together more to integrate specific needs of each group.
Street, C.T. (2006). Evolution in IS alignment and IS alignment capabilities over time: A test of punctuated equilibrium theory. Unpublished Doctor of Philosophy, Queen's University.	Case Study, $N=4$	The author examined the punctuated equilibrium model of organizational change. The author relied on the previous literature on alignment capabilities.	The goal of this thesis was to examine the process of alignment and see if there were cases where the punctuated equilibrium model did not hold. Secondly, this dissertation assessed how alignment capabilities change over time. Four key findings were reported: punctuated *vs*. gradualist alignment change; reinvention *vs*. thickening; positioning flows process; and fit *vs*. tension. The author provides a preliminary explanation of how the development of IT capabilities is associated with different IT alignment change patterns. In case studies, it was found that organizations go through four phases in alignment: punctuation, change, settling-in, and stability. The author found that in mature organizations, characterized by strong organizational inertia and institutionalism, seeking higher alignment may not be the most effective decision. Higher alignment is optimal up until a certain point, and then after that alignment may weaken an organization's ability to respond to certain threats. The author looked at how alignment capabilities developed over time and how they could be measured. In testing the punctuated equilibrium model, it was found that those organizations that followed the punctuated equilibrium model of alignment had their capabilities continually reinvented following the punctuated change. Those that did not follow the punctuated equilibrium model of change continually 'thickened' or strengthened their capabilities over time.
Tallon, P.P., Kraemer, K.L. and Gurbaxani, V. (2000). Executives' Perceptions of the Business Value of	Survey $N=304$ executives	In order to provide theoretical basis for their research model, the authors have examined the literature from	Using survey data from 304 business executives worldwide, it was found that corporate goals for IT can be classified into one of four types: unfocused,

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
Information Technology: A process-oriented approach, *Journal of Management Information Systems* **16**(4): 145.		two related perspectives. First, executives' perceptions as a proxy for realized IT business value. Second, a process-oriented approach to evaluating IT business value.	operations focus, market focus, and dual focus. The analysis confirms that these goals are useful indicators of payoffs from IT in that executives in firms with more focused goals for IT perceive greater payoffs from IT across the value chain. In addition, it is found that management practices such as strategic alignment and IT investment evaluation contribute to higher perceived levels of IT business value. Management practices play a central role in creating IT business value, helping to turn strategic intent for IT into position payoffs for the business. As corporations strive to realize greater payoffs from IT investments, in the short term this could be achieved through greater strategic alignment, reinforcing the need to specify clear and definitive goals for IT. In the longer term, corporations could consider moving toward more strategic or focused goals for IT, where IT payoffs can be considerably higher.
Tallon, P. (2003, November 15, 2003). Paul Tallon: The Alignment Paradox, *CIO Insight* **1**(47)	Survey $N=238$ midsize companies	This article investigates the alignment paradox, that is, while some companies benefit greatly from alignment, other do not, and in fact, suffer as a result of alignment.	IT alignment can sometimes fail to produce a business payoff. In a recent study of 238 midsize companies, it was not only found that this alignment paradox is possible, but sometimes strategic alignment might actually harm a company. The study found that while 70% of companies reduce costs or improve sales and customer service after increasing strategic alignment, 30% see no improvement or even a decline. The alignment paradox cannot be avoided just by picking certain technologies and avoiding others. Flexibility takes vigilance and smart management. IT executives must carefully size up the ability of each IT investment to either enhance flexibility or diminish it. Applying options-pricing models to your evaluation of an IT investment is one way to help you decide whether preserving flexibility for tomorrow is worth the cost today. Application service providers and outsourcing services provide companies with another way to achieve strategic alignment and IT flexibility at the same time. Outsourcing legacy systems can help a company to quickly get rid of inflexible, older systems. And as always, culture is important. There needs to be a mind-set that encourages shared networks and common IT procurement

Citation (Author, year)	Research method	Theory/concept	Findings
			policies, and an across-the-board willingness to give up best-of-breed systems that could be incompatible.
Tan, F.B. and Gallupe, B. (2006). Aligning Business and Information Systems Thinking: A Cognitive Approach, *Engineering Management, IEEE Transactions* **53**(2): 223–237.	Survey, $N=80$	This research is based on the notion of a shared domain of knowledge increasing alignment. The authors also draw on the Personal Construct Theory (Kelly, 1955) and the technique of cognitive mapping.	This paper examines alignment at its most micro-level – the cognitive level. The authors operationalize alignment as the shared cognition between business and IT executives. Shared cognition was used as a proxy measure for alignment. This was done using cognitive maps between business and IS executives. The results show that higher cognitive commonality is related to higher levels of alignment. That is, the higher the level of cognitive commonality between business and IT executives, the higher the levels of IT–business alignment. Similarly, greater diversity in cognitive structure and cognitive content of business and IT executives, the lower the expected levels of alignment. This has strong parallels with intellectual alignment and shared knowledge.
Tan, F. B. (1995). The Responsiveness of Information Technology to Business Strategy Formulation: An empirical study. *Journal of Information Technology* **10**(3): 171–178.	Survey, $N=88$ companies	The author utilizes the typology of business strategy (Miles and Snow, 1978) and the typology of IT-strategy responsiveness (Lucas and Turner, 1982; Johnston and Carrico, 1998; Venkatraman, 1991).	The goal of this research was to explore whether business strategy impacted the IT-strategy responsiveness in various organizations. Based on the strategic types developed by Miles and Snow (1978), the study reveals that variations in IT-strategy responsiveness are linked to the type of business strategy being pursued by the organization. IT is more responsive to business strategy in those organizations that emphasize innovation in their product and market strategies, as opposed to organizations operating in a relatively stable product/market area. The above conclusion implies that organizations with an aggressive business strategy are more likely to explicitly consider IT as a strategic resource in formulating business strategy. IT shapes a business strategy in a different extent, which differs among organizations and business strategy can help in understanding these differences.
Tan, F.B. (1997). Strategy Types, Information Technology and Performance: A study of executive perceptions. Paper presented at the	Survey, $N=65$ organizations	The author uses the strategy typology developed by Miles and Snow (1978). However, the focus in this paper is purely on prospectors and defenders	This case study reveals that fit between the firm's strategy and the use of IT is linked to a higher level of performance. The results are consistent with the notion of organizational fit. The study suggests that

Citation (Author, year)	Research method	Theory/concept	Findings
Proceedings of the Third AIS Americas Conference on Information Systems, Indianapolis, Indiana		since they differ in the ways that IT is utilized to deliver its strategy (Atkins, 1994; Tan, 1995).	prospector type organizations are more likely to use IT to expand product innovations and market opportunities, whereas defender type organizations are more likely to use IT to improve operational efficiency. The author argues that there is little perceived difference in performance – both prospector and defender type organizations reported generally favorable performance. The author suggests two possible scenarios, namely: • An organization's choice of strategy is a response to the process of adaptation to environmental forces and the type of performance, which is desired at the time. Defenders focus on improving operational efficiency to achieve excess profits or even a short-lived competitive advantage. Prospectors innovate in order to attain a better competitive position. • Since distinctive competencies exist, organizations achieve the desired results. However, prospectors, more than defenders, perceive marketing related competencies to be among their strengths.
Tavakolian, H. (1989). Linking the Information Technology Structure with Organizational Competitive Strategy: A survey, *MIS Quarterly* **13**(3): 309–317.	Survey, $N=52$ matched pairs from computer components industry	The author uses Miles and Snow's (1978) typology of strategic orientation (defenders have a conservative strategy whereas prospectors have an aggressive strategy and analyzers have a moderately competitive strategy). The author also specifies three types of IT activities: systems development and maintenance, systems operations, and systems administration.	This paper studies the fit between IT structure and the organizational competitive strategy. Specifically, the author is examining the relationship between strategic orientation and the centralization (or decentralization) of IT activities. The author hypothesized that defenders would have the most centralized IT activities, whereas prospectors would have the most decentralized. Degree of centralization was operationalized as the locus of responsibility. The findings indicate that there is a strong relationship between strategic orientation and IT structure, especially its degree of centralization. User departments in conservative (strategy) organizations have less control over their IT functions whereas user departments in more aggressive (strategy) organizations have much more control. The author recommends to practitioners that if one feels that they are managing an aggressive organization to gain control of the IT.

Citation (Author, year)	Research method	Theory/concept	Findings
Teo, Thompson S.H. and Ang, James S.K. (1999). Critical Success Factors in the Alignment of IS Plans with Business Plans, *International Journal of Information Management* **19**(1): 173–185.	Survey, $N=169$ senior IS executives	This study is an initial attempt to derive a set of critical success factors for aligning IS plans with business plans.	This article examines critical success factors (CSF) for aligning business plans with IS plans. The findings indicate that there are 12 success factors critical for aligning IS plans with business plans, with the top three being : • Top management commitment to the strategic use of IT is viewed as the most important factor. • IS management is knowledgeable about business. • Top management has confidence in the IS department. These CSFs are particularly useful for practitioners who are attempting to align their businesses. These CSFs provide practitioners with issues to focus on.
Thorogood, A., Yetton, P., Vlasic, A. and Spiller, J. (2003). Strategic IT at SA water: A case study in alignment, outsourcing and governance No. AT-0910-3). Australian Graduate School of Management: Cords/Welsearch Ltd and the Fujitsu Centre for Managing Information Technology in Organisations.	Case study, $N=1$ Australian Government-owned corp. SA Water	The South Australian Water case study illustrates the management challenges in aligning Information Technology with business objectives in a publicly owned corporation.	SA Water has been reorganized into a government-owned corporation with clearly set goals, including a focus on water quality. The information systems function fell behind industry standards but the CIO redirected it, refreshed the infrastructure and introduced the broker model of outsourcing. He introduced new roles and skills to match the strategy. The Information Services Executive Committee, Project Governance Office and Project Boards all helped to exercise control over information systems projects. The PRINCE2 methodology integrated these new roles and structures.
Van Der Zee, J.T.M. and De Jong, B. (1999). Alignment is Not Enough: Integrating business and information technology management with the balanced business scorecard, *Journal of Management Information Systems* **16**(2): 137–156.	Case study $N=2$ organizations	The authors elaborate on the concepts of the Balanced Scorecard (1991) in two case studies.	This paper addressed two main problems in business and IT management. The first problem was the time lag between business and IT planning processes. The second was the lack of common 'language' between business and IT management. The authors propose to use the Balanced Business Scorecard to tackle both problems. The benefits of using this approach, in contrast to traditional methods, are that business and IT management can use the same 'performance measurement' language, thereby integrating IT planning and evaluation fully into the business context. The Balanced Scorecard introduces overall goals and quantified norms for the whole business including IT. Secondly, integrating the business and IT management processes eliminates, or at least considerably reduces, the time lag between the two.

Citation (Author, year)	Research method	Theory/concept	Findings
Venkatraman, N. (1989). The Concept of Fit in Strategy Research: Toward Verbal and Statistical Correspondence, *Academy of Management Review* **14**(3): 423–444.	Conceptual	The author reviews and elaborates on the concept of fit citing disparities between the conceptualization of fit and the mathematical treatment of fit.	This article elaborates on the concept of fit and develops six fit perspectives: moderation, mediation, matching, gestalts, profile deviation, and covariation. Each of these six perspectives conceptualizes and measures fit in a particular way, with the conceptualization and calculation matching. The author stresses the importance of using the right conceptualization and measurement of fit because often times fit will be conceptualized one way and then measured using a different perspective. Researchers have used these terms interchangeably and this has led to weaknesses in the literature. Fit as moderation is conceptualized as a relationship between two variables, moderated by IT alignment. Fit as mediation is conceptualized as an intervention. IT alignment intervenes between one or more antecedents and the outcome variable. When using fit as mediation, the appropriate measurement scheme is path analysis. Fit as matching is conceptualized as a match between two variables. Fit as matching is measured using deviation score analysis, residual analysis, and analysis of variance. Fit as covariation is conceptualized as internal consistency among a set of related variables. Fit as covariation is measured using second-order factor analysis. Fit as profile deviation is conceptualized as the internal consistency of multiple contingencies. Fit as profile deviation is measured by using a sub-sample of high performers from a larger sample. The management profile of the high performers is estimated. Then the degree of adherence to this ideal profile by a given organization is obtained by calculating the Euclidean distance in an n-dimensional space. Fit as gestalts is conceptualized as fit as a pattern of relationships which are in a temporary state of balance. Fit as gestalts is measured using a cluster analysis and q-factor analysis.
Venkatraman, N., Henderson, J.C. and Oldach, S. (1993). Continuous Strategic Alignment: Exploiting Information Technology Capabilities for Competitive Success, *European Management Journal* **11**(2): 139–149.	Conceptual paper	The theoretical underpinnings of the Strategic Alignment Model developed in this paper are based on works of Henderson and Venkatraman (1991).	In this paper, the authors discuss analytical and administrative approaches for conceptualizing and managing the emerging nexus between strategic management and information technology. The analytical approach is discussed through the authors' Strategic Alignment Model – defined in terms of four

Citation (Author, year)	Research method	Theory/concept	Findings
			basic domains of strategic choice: business strategy, information technology strategy, organization infrastructure and processes, and information technology infrastructure and processes. The administrative approach for achieving strategic alignment is discussed in terms of a set of alignment mechanisms: governance process, technology capabilities, human resource capabilities, and value management.
Vitale, M.R., Ives, B. and Beath, C.M. (1986). Linking Information Technology and Corporate Strategy: An Organizational View. Proceedings of the 7th International Conference on Information Systems, San Diego, USA. 265–276.	Case study $N=17$ companies attending a conference	The authors present the top down model of strategy development (Chaffee, 1985; Chaandler, 1962), indicating its flaws. Then, the authors propose an exploratory question that for many organizations the environment will not support a top down process for identifying competitive information systems. The authors then introduce an adaptive model of strategy.	The authors identify three main reasons why the top down based methods for identifying strategic applications are unsatisfactory, namely (i) the overall strategy does not exist, (ii) strategies are uninformed about information systems technology, and (iii) environmental turbulence. The results reveal a strong relationship between IS knowledgeable management and system identification processes that are viewed as satisfactory. The survey validates the prediction that environmental turbulence associated with frequent changes in products, suppliers, customers, production processes, or competitive environment would make top-down planning more difficult. Moreover, such turbulence decreases the utility of the top-down planning process as an instrument for identifying competitive applications of information system technology. The authors conclude that development of an organizational structure conducive to the identification and implementation of strategic applications may prove to be sustainable competitive advantage.
Wang, E.T.G. and Tai, J.C.F. (2003). Factors Affecting Information Systems Planning Effectiveness: Organizational Contexts and Planning Systems Dimensions, *Information Management* **40**(4): 287–303.	Survey, $N=156$	This study is based on both the content and process dimensions of IS effectiveness. The conceptual framework is similar to that of Venkatraman and Ramanujam (1987). This framework states that organizational context impacts planning systems that in turn impacts IS planning system effectiveness.	This study examines the effects of organizational context and planning systems dimensions on the effectiveness of IS planning from a contingency perspective. The results demonstrate the pivotal role that an organization's planning capability plays in mediating the effects of organizational contexts and planning system dimensions of IS planning effectiveness. The authors looked at centralization and found no relationship with organizational alignment, a negative relationship with improvements in planning capability, and a positive relationship with

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
			environmental assessment. Thus, a centralized structure can weaken an organization's planning capabilities, but facilitates tasks such as environmental scanning. The authors also examined formalization and found a positive relationship with organizational coalignment, environmental assessment, and improvements in planning capabilities. The authors also found that for when the future role of IS in an organization was deemed to be important the organization paid a great deal of attention to the factors that facilitate IS planning and also to environmental assessment. The authors further found that environmental assessment improves the organization's planning process that in turn leads to greater alignment. The authors cite many implications from their research. Firstly, an organization's planning system is vital to achieving organizational alignment. Secondly, both the content and process of the planning system is important. Finally, organizational context should inform practitioners on what type of planning system to use. The authors suggest that future research should study this phenomena is a country other than Taiwan in order to improve generalizability.
Watson, R.T., Kelly, G.G., Galliers, R.D. and Brancheau, J.C. (1997). Key Issues in Information Systems Management: An international perspective, *Journal of Management Information Systems* **13**(4): 91–115.	Secondary Data Analysis	This study compares and contrasts the findings of recent information systems (IS) management studies in ten nations or regions as well as one US multinational study. The studies – of Australia, Estonia, Europe, the Gulf Cooperative Council, Hong Kong, India, Slovenia, Taiwan, UK, and US – selected for analysis represent the most current and consistent data available.	This study examines the key concerns of IS executives in these different geographical areas, focusing on identifying and explaining regional similarities and differences. Internationally, there are substantial differences in key issues. Differences leading to key issues in different geographic locations include cultural, economic development, political/legal environment, and technological status factors. The analysis suggests that national culture and economic development can explain differences in key issues. The paper concludes with a revised framework for key issues studies that will more readily support comparison across time and nations. Nations are classified in terms of the power distance and uncertainty avoidance. Low power distance and low uncertainty avoidance is labeled as 'village market development' and includes countries such as Australia, the United Kingdom, and the United States. High

Citation (Author, year)	Research method	Theory/concept	Findings
			power distance and low uncertainty avoidance is labeled as 'family development' and includes countries such as Hong Kong. Low power distance and high uncertainty avoidance is labeled as a 'well-oiled machine' and no countries in the study fit this profile. Finally, high power distance and high uncertainty avoidance is labeled as a 'pyramid of people' and includes countries such as the Gulf cooperative and Slovenia. This research suggests that IT managers moving from one geographic location to another need to be mindful of these differences.
Wehmeyer, K. (2005). Aligning IT and Marketing – The Impact of Database Marketing and CRM, *Journal of Database Marketing & Customer Strategy Management* **12**(3): 243–256.	Case Study, $N=2$	The author uses the strategic alignment model by Henderson and Venkratraman (1993). The authors review the differences between database marketing (IT-enhanced direct marketing) and customer relationship management (transactional marketing-mix marketing by direct means).	This article investigates the usefulness of the SAM model in the context of database marketing and consumer relationship management (CRM). Discussing database marketing and CRM in terms of strategic IT adds to the discussion about the differences between the two marketing methods. SAM demonstrates the critical role of IT and IT executives in marketing efforts. Secondly the SAM can be used as a task analysis for marketing efforts.
Whittington, R. (ed.). (1993). *What is Strategy – and Does it Matter. Routledge Series in Analytical Management*, London: Routledge.	Conceptual	The author reviews the various theories of strategy.	This book attempts to explain strategy and its impact. More importantly, the author outlines the various theories of strategy: classical, evolutionary, processual, and systemic. The classical approach to strategy is also known as the rational approach. This perspective emphasizes formal, explicit, long-term plans from the top down. The classical approach relies heavily on economic theory. The evolutionary perspective relies on the markets and views strategy as less rational and more as 'survival of the fittest'. Survival in the business world – as in the natural world – comes down to differentiation. Two companies that are exactly the same cannot survive in the same market. The evolutionary perspective is more short-term focused and advocates the market – not the manager – choosing what strategy is the best. The processual school is also skeptical about the rationality of managers, but does not trust the markets as much as the evolutionary theory. The processual school advocates dealing with the situation as it presents itself. Fundamental to this perspective are the cognitive

Citation (Author, year)	Research method	Theory/concept	Findings
			limits on rational action and the micro-politics of an organization. Finally, the systemic approach is discussed. The systemic school advocates for a long-term management approach similar to the classical school, but where it differs is in the fact that it recognizes the sociological context where these strategies and decisions are formulated. The actions and ends of strategies are context-specific and some organizations function in contexts that are not perfectly profit-maximizing. In fact, the notion of 'strategy' may be considered contextually dependent.
Yetton, P.W. and Sauer, C. (1997). The Paths Ahead, in C. Sauer and P.W. Yetton (eds.) *Steps to the Future: Fresh Thinking on the Management of IT-Based Organizational Transformation*, San Francisco: Jossey-Bass ed., pp. 279–303.	Literature Review and Conceptual Paper	This is a concluding chapter of a book, summarizing previous chapters and looking ahead.	At the present time organizations are recognized as consisting of multiple constituencies, whose competitive interactions yield unpredictable outcomes, and IT is just one element of the organization that interacts with others. Redirection of management attention reduces the problem of the separation of business and IT. IT is employed in gaining achievable business benefits, business managers together with IT professional get closer than they have been and discover shared interests and opportunities. The problem of alignment disappears and IT becomes more manageable. Looking to the future, the authors state that as organizations become more complex, the need to understand the under-lying intricacies will become ever more important.
Yetton, P.W. (1994). False Prophecies, Successful Practice and Future Directions in IT Management. Proceedings of the IFIP TC8 Open Conference: Business Process Re-Engineering: Information Systems Opportunities and Challenges, Queensland Gold Coast, Australia.	Case study $N=3$ Australian organizations	The author utilizes the MIT 90 s framework of strategy-structure fit (Scott Morton, 1991) and the conventional model of strategic dynamics (Yetton and Johnston, 1993)	This research attempts to provide insight into the benefits of the IT revolution. The author concludes that if the separation between IT and the business is substantial, company performance will suffer. Separation also contributes to the failure of organizations to learn how to manage IT investments. Dynamics, which create a gulf between business-illiterate IT groups and IT-illiterate top management, make it very difficult to assimilate understanding required to add value to each of these groups. Strategic alignment should seek a balance between (i) centralization and devolution, and (ii) IT strategies that focus on cost effectiveness and corporate standards, and those that are value-added and business driven. This paper also raises the possibility that

Citation (Author, year)	Research method	Theory/concept	Findings
			alignment can sometimes best be achieved via outsourcing. The study contradicts the prevailing preoccupation with aligning the IT strategy to the business strategy by showing that the process by which the service is delivered to a customer has been changed, in the sense of becoming more integrated and simplified, i.e. re-engineered. The author leaves researchers and practitioners with a few key take-aways. Namely, the author states the researchers will have difficulty producing large-scale models and instead should focus on the anomalies, where companies had to do things a little differently. For practitioners, the author suggests that IT should no longer be the 'out there' function and it should be embraced into business functions.
Yetton, P.W., Johnston, K.D. and Craig, J.F. (1994). Computer-Aided Architects: A case study of IT and strategic change, *Sloan Management Review* **35**(4): 57–67.	Case study $N=1$ organization	Although the authors used the MIT model (Scott Morton, 1991), the authors describe and analyze a case in which business transformation occurred along a different – almost reverse to the conventional, rational models in the MIT Management in the 1990 s framework – path to fit, through the incremental adoption of IT.	This research looks at a case study where alignment was achieved in a way counter-intuitive to traditional alignment models. This case study provides several insights into the dynamic nature of IT change in organizations, some of which challenge conventional wisdom. Business transformation emerged from what was essentially a series of tactical, rather than strategic, decisions. The change process occurred incrementally, on a project-by-project basis, reducing business and financial risk, while standardization on a common technology platform maximized accumulated learning over time. The continual adaptation of technology through individual mastery and organizational learning was central to achieving a strategic fit in which IT became embedded in the firm's core business process, contributing to the firm's high performance and the development of competitive advantage. Practicing professionals, rather than IT specialists, led the move to adopt and implement technology. Their early commitment to mastering the technology meant that it enhanced and empowered the professional role, rather than subordinating or replacing it.
Yetton, P.W., Craig, J.F. and Johnston, K.D. (1995). Fit, Simplicity and Risk: Multiple Paths to Strategic	Case study, $N=3$ organizations	This paper develops a theoretical basis for defining alternative paths to IT-centered strategic change that are	This research explores the dynamic and multiple paths to fit. Three cases are used to show that, in practice, there are multiple alternative paths to successful

Citation (Author, year)	*Research method*	*Theory/concept*	*Findings*
IT Change. Proceedings of the Sixteenth International Conference of Information Systems (ICIS'95), Amsterdam, The Netherlands. 1–11.		robust and sustainable, from three literatures: 1. Strategic IT literature 2. Dominant strategy literature 3. Organization theory contingency literature on fit, which underpins the MIT 90 s model.	organizational transformation. The strategic IT and business strategy literatures are critiqued in terms of the process of fit so as to explain why these alternative paths are successful. This paper identified three additional dynamic paths to fit in addition to the dominant conventional one. In particular, the authors argue that the three case studies have in common the search for simplicity, but are characterized by different technological and business risks (risk management profiles), which may be located in a step or a property of the path. This paper provides the implication for researchers that the multiple paths to achieving fit require different analyses and contingencies.
Zviran, M. (1990). Relationships Between Organizational and Information Systems Objectives: Some empirical evidence, *Journal of Management Information Systems* 7(1): 65–84.	Survey, $N = 131$ (Israel)	The author reviews the strategic planning literature and outlines for organizations to develop their objectives.	This paper presents an empirical study looking at organizational and IS objectives. Their overall hypothesis is that organizational and IS objectives must be linked. The results represent the first empirical work to support this hypothesis. The following organizational objectives were studied: control and reduce costs, increase revenue, improve administrative efficiency, improve services, supply products and services on time, gain competitive advantage, improve product quality, and increase organizational productivity. Results support a relationship between organizational and IS objectives.

About the authors

Yolande E. Chan is a Professor and E. Marie Shantz Research Fellow in MIS at Queen's University in Canada. She holds a Ph.D. from the University of Western Ontario, an M.Phil. in Management Studies from Oxford University, and S.M. and S.B. degrees in Electrical Engineering and Computer Science from MIT. Prior to joining Queen's, she worked with Andersen Consulting (now Accenture). Currently she serves as Director, The Monieson Centre. Dr. Chan conducts research on information systems alignment, knowledge management, information privacy, and information systems performance. She has published her findings in journals such as *Information Systems Research, MIS Quarterly Executive, Journal of Management Information Systems, Journal of the Association for Information Systems, Journal of Strategic Information Systems, Information & Management*, *IEEE Transactions on Engineering Management* and *The Academy of Management Executive*. Dr. Chan is a member of several journal editorial boards and is an officer of the Association for Information Systems.

Blaize Horner Reich is a Professor at Segal Graduate School of Business, Simon Fraser University, Vancouver, Canada.and is a visitingaAssociate at Templeton College, Oxford University. Before pursuing her Ph.D., Dr. Reich worked for 15 years in the IT industry as a practitioner and consultant, focusing on the financial services and utilities sectors. She is currently an editorial board member for several academic journals and a corporate director of two companies.

Dr. Reich's research has been published in a wide range of journals, including *MIS Quarterly, Information Systems Research, Journal for Management Information Systems and the Project Management Journal*. Dr. Reich has two current research programs: (1) IT Project Performance and (2) IT Governance and Risk management. She and colleagues at Oxford and SFU recently received two national grants to extend their IT Project Performance research.